

# Indexed Bellman Information Complexity

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## Abstract

We develop *indexed Bellman information complexity*, a representation-level framework for reinforcement learning and interactive decision making. The primitive object is an indexed sequential learning with a full environment  $\Omega$  and admissible nonanticipating policies; priors, posteriors, and ghost histories are used as reference or certificate devices for the fixed frequentist truth. The proof object is a Bellman-sufficient state together with an information index  $Y = \chi(\Omega)$ , often the optimal decision or value object rather than the full environment. The state must preserve the losses, predictive laws, reference updates, and indexed log-gain increments needed for Bellman recursion; this sufficiency condition is the representation-theoretic replacement for a purely syntactic model-realizability assumption. On the upper-bound side, the learning problem can be organized around a dynamic program with a logarithmic information potential. In fixed-truth analysis this potential is represented by the coordinate log loss  $\gamma \log(1/q_t(\chi(\omega^*)))$ ; in the indexed Algorithmic Information Ratio (AIR) regret identities it gives rise to the log-posterior telescope, and after Bayesian posterior averaging it corresponds to an entropy term. On the lower side, a Bellman-Fano certificate uses the same state and index to compare the indexed information telescope with the ghost-good mass of low-regret reference trajectories. The central matching statement is therefore a conditional Bellman information-risk sandwich when the log-penalized Bellman upper value and the ghost-quantile lower certificate close on the same representation and at the same radius. UCB, E2D/DEC, and AMS/EBO then appear as tractable certificates or relaxations of this same log-potential Bellman program, rather than as separate notions of information complexity.

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# 1 Introduction

Information-theoretic minimax theory balances two quantities: the information revealed by an experiment and the logarithmic mass of the set on which an estimator can be successful. In ordinary, noninteractive statistical estimation the experiment is fixed in advance, so this balance can be expressed through local entropy, prior mass, and KL information; this is the perspective behind the information-risk upper and lower bounds of [Zhang \(2006\)](#) and the entropy characterization of minimax rates in [Yang and Barron \(1999\)](#), together with classical Fano and Assouad methods ([Yu, 1997](#)). In interactive decision making the learner designs the experiment while learning. The action law at round  $t$  is a statistic of the past observations, so it may itself reveal information about the unknown environment. A fixed-design information calculation is therefore not enough. This adaptivity is exactly what makes a sharp theory difficult.

A powerful response is algorithmic Fano’s method, meaning a Fano-type inequality in which the comparison data set is generated from an algorithm-dependent reference or “ghost” experiment rather than from a fixed nonadaptive sampling scheme. This perspective appears in recent interactive Fano frameworks for general decision making ([Chen et al., 2024](#)) and in algorithmic lower bounds for statistical learning ([Xu, 2026](#)); it is also closely related to the decision-estimation coefficient (DEC) approach of [Foster et al. \(2021\)](#) and its constrained/localized refinements ([Foster et al., 2023](#)). The central obstacle in the algorithmic Fano method is what comes next: collapsing a  $T$ -round adaptive experiment into a single one-round action distribution generally requires a localized change-of-measure argument, because the action law under a hard environment need not coincide with the action law under the reference experiment. Such a reduction can obscure or discard the geometry of the posterior trajectory, making the theory more rigid than necessary.

The thesis of this paper is that the right invariant object is not a one-shot coefficient by itself, nor the full environment specification by itself, but a Bellman-sufficient representation together with an index. The full environment  $\Omega$  remains the object that generates rewards and observations. The state records the part of the history needed to continue the Bellman recursion. The index  $Y = \chi(\Omega)$  records the information whose acquisition is charged, such as the optimal action, optimal policy,

active reward vector, or full model. In favorable problems the state is a minimal or nearly minimal sufficient statistic, such as a posterior, a finite marginal posterior, a calibrated confidence state, or a Bellman-error witness representation. When such a state is found, learning becomes a special dynamic program whose generic value-like potential is logarithmic information about the index. This differs from ordinary computational dynamic programming, where the value function and algorithm are usually identified case by case for a particular model; here the same entropy, relative-entropy, and log-partition potentials generate broad algorithmic families.

The basic upper-side identity is fixed-truth and coordinatewise. Fix a truth  $\omega^*$ , let  $y^* = \chi(\omega^*)$ , and let  $q_t$  be the retained reference marginal on the index. The specialized logarithmic potential

$$\phi_t(s; y^*) = \eta^{-1}[-\log q_t(y^*)]$$

turns the one-step fixed-truth log gain into a Bellman potential drop. Thus the AIR/MAIR bracket is

$$\ell_{\omega^*}(S_t, p_t) + \mathbb{E}_{\omega^*}[\phi_{t+1}(S_{t+1}; y^*) \mid S_t, p_t] - \phi_t(S_t; y^*),$$

and its sum telescopes to a regret bound of order  $\eta^{-1} \log(1/q_1(y^*))$  plus the accumulated bracket errors. Bayesian posterior averaging of the same coordinate identity gives the chain rule

$$I_\mu(Y; H_T) = \mathbb{E}_{H'_T} \sum_{t=1}^T \mathcal{I}_\chi(S'_t, p'_t),$$

where  $S'_t$  is the reference posterior state and  $p'_t$  is the algorithm's action distribution at that reference history.

The same reference history also defines the ghost probability that an algorithm has small target-wise average regret. The lower-bound theorem takes the form

$$\text{if } T\bar{C}_\chi^{\text{Alg}}(\mu) \leq \text{kl}\left(\frac{1}{2}, p_r^{\text{Alg}}(\mu, \chi)\right), \quad \text{then } \mathbb{E}_\mu^{\text{Alg}} \text{Reg}_T \geq \frac{Tr}{2}.$$

Here  $r$  is an average-regret threshold,  $p_r^{\text{Alg}}$  is the posterior-reference ghost probability of average regret at most  $r$ , and  $\bar{C}_\chi^{\text{Alg}} = T^{-1} \mathbb{E} \sum_t \mathcal{I}_\chi(S'_t, p'_t)$  is average indexed information. The critical balance is

$$T\bar{C}_\chi \asymp \log \frac{1}{p_r}.$$

When the ghost-good mass is roughly  $\exp(-d_{\text{eff}})$ , the admissible total information is  $d_{\text{eff}}$ -level. This is the Fano/local-entropy radius, not the constant two-point radius.

Theorem 4.4 turns this balance into a frequentist minimax statement. After a state, index, reference update, and calibration mechanism have been fixed, the upper side is a log-penalized Bellman supersolution with coordinate cost  $\gamma \log(1/q_1(\chi(\omega)))$ ; the lower side is a Bellman-Fano ghost certificate with entropy  $-\log p_r$ . If these two quantities meet at the same radius, the statistical part of the interactive decision problem is solved at that radius. This is the precise interactive analogue of noninteractive information-risk matching: the fixed experiment is replaced by a controlled Bellman recursion, and local prior mass is replaced by ghost-good mass along the reference history.

After fixing a prior  $\mu$ , an index  $\chi$ , and a state map  $S_t = \phi_t(H_{t-1}, b_t)$ , the two sides are comparable only if they use the same representation. In exact Bayes analyses the state contains the true posterior  $b_t$ ; in frequentist or UCB-style analyses it may contain an algorithmic belief or confidence object, but then a separate calibration step must relate that object to the fixed truth. The

upper side supplies either the fixed-truth log-potential certificate of Theorem 4.1, the calibrated frequentist log-penalized Bellman program of Theorem 4.2, or the Bayesian information-potential Bellman program of Theorem 4.3. The lower side supplies a Bellman-Fano certificate: an information supersolution and a ghost-good-mass certificate, as in Theorem 5.3, built on that same chosen state process and the same index. We use “indexed Bellman information complexity” for this compatible state/index accounting. It becomes a matching complexity statement only after the upper and lower certificates are separately verified and shown to agree up to the stated constants or logarithmic factors.

The word “Bellman” in the title is deliberate. The exact value of an interactive problem is a dynamic program, following Bellman’s principle of optimality (Bellman, 1957). An upper bound is a Bellman supersolution, or admissible relaxation in the sense of Rakhlin et al. (2012): it lies above the dynamic value and gives an algorithm. A lower bound is a Bellman subsolution or, more flexibly, a Bellman-Fano certificate that combines an information supersolution with a ghost-good-mass supersolution. The canonical information-theoretic upper algorithm is therefore the AIR/MAIR information-potential Bellman program itself: it keeps the continuation value on the indexed state and optimizes the Bellman bracket before any one-step relaxation is imposed. UCB, DEC/E2D, and AMS/EBO are useful because they certify, relax, or robustify this Bellman bracket; they should not be confused with the exact Bellman program itself.

## 1.1 Main contributions.

1. We formulate *indexed sequential learning*, a representation-level framework for reinforcement learning and interactive decision making in which the problem is to choose an admissible policy satisfying Bellman optimality, while a chosen index  $Y = \chi(\Omega)$  records the information whose acquisition is charged. The state must be sufficient for losses, observation kernels, posterior predictive laws, posterior updates, and indexed information increments. This subsumes DMSO as a special case (Foster et al., 2021), but the primitive condition is Bellman sufficiency rather than membership in a syntactic model class.
2. We prove the indexed posterior-reference chain rule and the AIR/MAIR regret identities. AIR corresponds to the decision index  $Y = \pi^*(\Omega)$ ; MAIR corresponds to the model or environment index  $Y = \Omega$ . We record simplifications and specializations, the gradient brackets, and the reason Danskin’s theorem appears in the original AIR proof of Xu and Zeevi (2025).
3. We give a unified “one identity, four algorithm families” account centered on a log-penalized Bellman upper theorem and an information-risk sandwich. The first family is the exact AIR/MAIR information-potential Bellman program, which optimizes the dynamic bracket on the chosen state and is the upper object that can match the Bellman-Fano lower certificate. UCB methods (Auer et al., 2002; Abbasi-Yadkori et al., 2011; Srinivas et al., 2010) certify the fixed-truth log-potential bracket through self-normalized calibration and optimism. E2D/DEC methods (Foster et al., 2021, 2023) optimize a one-step offset obtained by replacing the continuation value by a local statistical separation penalty. AMS/EBO methods (Lattimore and György, 2021; Xu and Zeevi, 2025) optimize KL-regularized robust belief relaxations, equivalently dual log-partition relaxations, of the same indexed Bellman bracket.
4. We prove the reference-history quantile lower theorem and Bellman lower certificates. As an indexed posterior-reference specialization of the interactive Fano method, the lower side uses the same stepwise indexed information as the upper side, but compares it to ghost-good entropy rather than to a one-round coefficient. Through bandit examples we show DEC/E2D

is best viewed as a one-step relaxation of indexed Bellman information complexity, not as a universally tight conversion coefficient.

5. We develop several representative examples, including kernel bandits with an indexable action set, multi-armed bandits, linear bandits, and a Bellman-rank embedding. The kernel-bandit example highlights a regime where the optimal active action is finite, even though the unknown RKHS function remains infinite-dimensional. This makes indexed AIR or optimism the natural theory, while global DEC may be vacuous unless it is properly localized.

## 1.2 Organization.

Section 2 defines indexed sequential learning and Bellman-sufficient representation. Section 3 gives the indexed information telescope and the exact AIR/MAIR regret identities. Section 4 the log-penalized Bellman upper theorem, the information-risk matching principle, and upper algorithmic relaxations including UCB families, E2D/DEC, and AMS/EBO. Section 5 gives the quantile lower theorem, Bellman-Fano certificates, and the status of one-round DEC. Section 6 gives applications to kernel bandits, MAB, linear bandits, and Bellman-rank embedding.

# 2 Indexed sequential learning and Bellman-sufficient representation

## 2.1 Frequentist dynamic programming before representation

The primitive problem is frequentist. There is an environment class  $\Omega$ , and performance is required for each fixed truth  $\omega^* \in \Omega$ . A prior  $\mu$  is introduced only when we form a Bayes value, a Yao lower bound, a posterior-reference trajectory, or an algorithmic belief. At time  $t$ , after the history

$$H_{t-1} = (A_1, O_1, \dots, A_{t-1}, O_{t-1}),$$

the learner chooses a distribution  $p_t(\cdot \mid H_{t-1})$  over available actions and samples  $A_t$  from it. Under truth  $\omega^*$ , the observation is drawn from

$$O_t \sim P_{\omega^*, t}(\cdot \mid H_{t-1}, A_t),$$

and the one-step loss or regret is  $\ell_t(\omega^*, H_{t-1}, A_t)$ . The exact frequentist objective is

$$\mathfrak{R}_T^*(\Omega) := \inf_{\text{Alg}} \sup_{\omega \in \Omega} \mathbb{E}_{\omega}^{\text{Alg}} \sum_{t=1}^T \ell_t(\omega, H_{t-1}, A_t).$$

This is the reinforcement-learning problem in its most general controlled-experiment form: choose an admissible nonanticipating policy whose Bellman value is small for every environment.

The full history is always a valid Bellman state. A compressed state is legitimate only when it preserves the objects needed for the Bellman recursion. Let  $S_t$  be such a state, with action set  $\mathcal{A}(S_t)$ , transition/update map  $S_{t+1} = \tau_t(S_t, A_t, O_t)$ , and state-dependent kernels  $P_{\omega, S_t, A_t}$ . If a Bayesian mixture law with posterior  $b_t = \mathcal{L}_{\mu}(\Omega \mid H_{t-1})$  is used as a reference, the exact Bayes-regret dynamic program on an exact sufficient state is

$$V_{T+1}(s) = 0, \quad V_t(s) = \inf_{p \in \Delta(\mathcal{A}(s))} \left\{ \ell(s, p) + \mathbb{E}_{a \sim p, o \sim P_{s, a}} V_{t+1}(\tau(s, a, o)) \right\}, \quad (1)$$

where

$$\ell(s, p) = \mathbb{E}_{a \sim p} \int \ell_\omega(s, a) b_s(d\omega), \quad P_{s,a} = \int P_{\omega,s,a} b_s(d\omega).$$

Equation (1) is not the definition of the frequentist problem; it is the Bayes/reference Bellman recursion induced by the chosen representation. Minimax lower bounds are connected to it through Yao's principle (Yao, 1977),

$$\mathfrak{R}_T^*(\Omega) \geq \sup_{\mu \in \Delta(\Omega)} \inf_{\text{Alg}} \mathbb{E}_{\Omega \sim \mu} \mathbb{E}_\Omega^{\text{Alg}} L_\Omega(H_T).$$

The information-complexity layer asks whether the same frequentist problem can be controlled after compressing the history to a Bellman-sufficient state and charging information only about an index  $Y = \chi(\Omega)$ . The state should be as small as the proof permits—minimal when a minimal sufficient statistic exists, otherwise simply sufficient. The index is the final coordinate whose logarithmic mass or ghost entropy controls the statistical price of learning. This is the sense in which learning is a special dynamic program: unlike ordinary computational dynamic programming, where the value function is problem-specific, the statistical potential is generically logarithmic in the index belief.

**Definition 2.1** (Bellman supersolution and subsolution). *Let  $\mathsf{T}_t$  be the Bellman operator in (1). A sequence  $\bar{V}_t$  is a regret supersolution if*

$$\bar{V}_{T+1} \geq 0, \quad \bar{V}_t(s) \geq (\mathsf{T}_t \bar{V}_{t+1})(s) \quad \text{for all } t, s.$$

*A sequence  $\underline{V}_t$  is a regret subsolution if*

$$\underline{V}_{T+1} \leq 0, \quad \underline{V}_t(s) \leq (\mathsf{T}_t \underline{V}_{t+1})(s) \quad \text{for all } t, s.$$

**Proposition 2.2** (Bellman sandwich). *If  $\bar{V}$  is a supersolution, then any policy that attains the right-hand side of the supersolution inequality up to errors  $\varepsilon_t$  has Bayes regret at most  $\bar{V}_1(s_1) + \sum_t \varepsilon_t$ . If  $\underline{V}$  is a subsolution, then every algorithm has Bayes regret at least  $\underline{V}_1(s_1)$ . Consequently,*

$$\underline{V}_1(s_1) \leq V_1(s_1) \leq \bar{V}_1(s_1).$$

*Proof.* For the upper statement, if the selected decision kernel satisfies

$$\ell(s, p_t) + \mathbb{E} \bar{V}_{t+1}(S_{t+1}) \leq \bar{V}_t(s) + \varepsilon_t,$$

then summing over  $t$  telescopes the potential and yields the claimed bound. The lower statement is the same induction with the inequality reversed: for any decision kernel  $p_t$ ,

$$\ell(s, p_t) + \mathbb{E} \underline{V}_{t+1}(S_{t+1}) \geq \underline{V}_t(s).$$

Summing gives expected loss at least  $\underline{V}_1(s_1) - \mathbb{E} \underline{V}_{T+1}(S_{T+1}) \geq \underline{V}_1(s_1)$ .  $\square$

This proposition is the organizing principle of the paper. The upper theory constructs supersolutions by paying the Bellman bracket with information. The lower theory constructs subsolutions indirectly, by showing that any algorithm with too little information has too small a ghost-good probability to achieve low regret. The representation required by both sides is exactly the representation in which these Bellman backups are well-defined.

## 2.2 Latent environments, indices, and losses

The latent environment space is denoted by  $\Omega$ . An information index is a measurable map

$$Y = \chi(\Omega) \in \mathcal{Y}.$$

The model-index choice  $Y = \Omega$  gives MAIR. The action-index choice  $Y = A^*(\Omega)$ , where  $A^*$  is an optimal action under the environment, gives AIR. The loss may depend on more than  $Y$ . For index-based lower bounds we therefore use the conditional loss

$$\ell_t^X(y, H_{t-1}, a) := \mathbb{E}_\mu[\ell_t(\Omega, H_{t-1}, a) \mid \chi(\Omega) = y, H_{t-1}], \quad (2)$$

assuming the relevant regular conditional distributions exist. Define

$$L_\omega(H_T) = \sum_{t=1}^T \ell_t(\omega, H_{t-1}, A_t), \quad \bar{L}_\omega(H_T) = T^{-1} L_\omega(H_T),$$

and

$$L_\chi(y, H_T) = \sum_{t=1}^T \ell_t^X(y, H_{t-1}, A_t), \quad \bar{L}_\chi(y, H_T) = T^{-1} L_\chi(y, H_T).$$

Under the true Bayesian experiment,

$$\mathbb{E}_{\Omega \sim \mu, H_T \sim \mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) = \mathbb{E}_{Y, H_T} L_\chi(Y, H_T). \quad (3)$$

The choice of information index is fairly flexible. If a pathwise, model-wise event is needed, one may take  $Y = \Omega$ . If the objective is decision identification, one may take  $Y = A^*(\Omega)$  and pay only for information about the optimal decision. If a value-function class  $\mathcal{F}$  provides Bellman sufficiency under the chosen state representation, one may take  $Y = f^*(\Omega)$  and pay only for information about the optimal value function.

## 2.3 Bellman-sufficient representation

We work with an indexed sequential experiment. The environment  $\omega \in \Omega$  is fixed in the frequentist problem. A prior  $\mu$  over  $\Omega$  is used when we form Bayes values, posterior-reference trajectories, algorithmic beliefs, or Yao/Fano lower bounds. At round  $t$ , the learner observes a Bellman state  $S_t$ , chooses  $A_t \in \mathcal{A}(S_t)$ , observes  $O_t \sim P_{\omega,t}(\cdot \mid S_t, A_t)$ , and incurs a nonnegative per-round regret/loss  $\ell_\omega(S_t, A_t) \in [0, L]$  (for upper-bound analyses with misspecification or adversarial losses, one may use a nonnegative surrogate loss). The next state is  $S_{t+1} = \tau_t(S_t, A_t, O_t)$ .

The full history is always an admissible state. A compressed state is valid only when it is Bellman sufficient: it preserves exactly the objects needed for future prediction, loss evaluation, updating, and indexed information accounting. Thus the representation is the replacement for syntactic model realizability. One need not estimate the whole environment if a smaller state closes the Bellman and information recursions. In Bayesian bandits this state may be the posterior over models. In UCB analyses it may be an estimator together with a confidence set or covariance matrix. In finite-action kernel bandits it is the finite active marginal posterior. In low-Bellman-rank contextual decision processes it must include the witness, value-function, and roll-in information needed to predict Bellman errors; Bellman rank alone is not a state.

The framework separates the statistical experiment from the proof device. The minimax objective is frequentist: the environment  $\omega$  is fixed. Priors and posteriors are used either as algorithmic

beliefs or as reference measures for Bayes/Yao/Fano arguments. A Bayesian upper bound becomes a frequentist upper bound only after a calibration or domination step showing that the algorithmic belief controls the fixed truth. We do not assume that a unique smallest sufficient state always exists. When a minimal sufficient representation is available it is the natural state; otherwise any exact or approximate Bellman-sufficient representation may be used, with approximation errors inserted into the Bellman inequalities. Algorithmic beliefs and posteriors, DMSO estimators, UCB confidence states, and Bellman-rank witnesses are therefore different representations of the same underlying indexed sequential experiment.

**Definition 2.3** (Indexed Bellman-sufficient representation). *Fix a prior  $\mu$  and an index map  $\chi$ . Let  $b_t = \mathcal{L}_\mu(\Omega \mid H_{t-1})$  be the posterior under the Bayesian mixture law. A representation is a process*

$$S_t = \phi_t(H_{t-1}, b_t).$$

*The time index may be included in  $S_t$ ; equivalently, the kernels and updates may be time-inhomogeneous, and we suppress the subscript  $t$  when it is part of the state. It is exact Bellman-sufficient if, for every state  $s$ , action  $a \in \mathcal{A}(s)$ , and environment  $\omega$ , there exist objects*

$$P_{\omega,s,a} \in \Delta(\mathcal{O}), \quad \ell_\omega(s, a) \in [0, L], \quad \tau(s, a, o),$$

*such that the following hold.*

- (i) Predictive sufficiency:  $P_{\omega,t}(\cdot \mid H_{t-1}, a) = P_{\omega,S_t,a}$ .
- (ii) Loss sufficiency:  $\ell_t(\omega, H_{t-1}, a) = \ell_\omega(S_t, a)$ .
- (iii) Posterior predictive sufficiency:  $P_{s,a} := \int P_{\omega,s,a} b_s(d\omega)$  where  $b_s := \mathcal{L}_\mu(\Omega \mid S_t = s)$ , is determined by  $s$ .
- (iv) Update sufficiency: under the posterior-reference law,  $S_{t+1} = \tau(S_t, A_t, O_t)$ .
- (v) Index sufficiency: for  $q_s = \mathcal{L}(\chi(\Omega) \mid S_t = s)$  and  $b_s^y = \mathcal{L}(\Omega \mid S_t = s, \chi(\Omega) = y)$  defined for  $q_s$ -almost every  $y$ , the conditional predictive laws

$$P_{s,a}^y := \int P_{\omega,s,a} b_s^y(d\omega)$$

*and conditional losses*

$$\ell_y^x(s, a) := \int \ell_\omega(s, a) b_s^y(d\omega)$$

*are determined by  $s$ .*

The state should be understood as a controlled sufficient statistic for both information accounting and algorithm design. Classical sufficiency starts from a statistic  $T$ , a random variable constructed from the data  $X$  alone and not depending on the unknown parameter  $\theta$ . The statistic  $T$  is sufficient for the model  $\{P_\theta : \theta \in \Theta\}$  if, after conditioning on  $T$ , the remaining conditional distribution of the data  $X$  is independent of  $\theta$  (Fisher, 1922). Here the analogue is dynamic and decision-theoretic: a state is sufficient if it preserves everything needed for future prediction, loss, information, and updating under admissible controls.

The index  $Y = \chi(\Omega)$  is not required to identify the whole environment. It is the coordinate whose posterior log mass is charged by the information potential and whose ghost-good mass appears in the Fano certificate. A posterior is sufficient in ordinary Bayesian bandits. A Gaussian



mean-covariance pair is sufficient for finite-dimensional Gaussian linear bandits. In finite-action kernel bandits, the active marginal posterior mean and covariance on the active action set are sufficient for the finite experiment, even though the unknown function remains an RKHS element. For low Bellman-rank contextual decision processes, a valid state must include the witness features, value-function class, roll-in distribution, and any posterior or confidence object needed to predict future Bellman errors; bounded Bellman rank alone is not a model class.

DMSO (Foster et al., 2021) is one instantiation:  $\Omega$  is a model  $M$ , an action is a decision  $\pi$ ,  $P_{M,\pi}$  is the observation law, and the state may be an estimator, confidence set, Gaussian belief, or other calibrated representation. However, the DMSO formalism itself is not required for, nor intrinsic to, tight upper and lower bounds in indexed sequential experiments. The indexed Bellman framework in Definition 2.3 is more primitive. Its true requirement is Bellman sufficiency: whether the chosen state is sufficient for prediction, updating, loss evaluation, and information accounting. Approximate representations can be used by adding explicit approximation errors to the Bellman inequalities.

This is aligned with the representational philosophy behind low Bellman rank for contextual decision processes (Jiang et al., 2017). In particular, contextual decision processes with low Bellman rank provide an important example of representation-level structure. Bellman rank is not a posterior over models; it is a factorization of Bellman errors relative to a chosen function class, roll-in policy family, and sampling scheme. Therefore bounded Bellman rank is not, by itself, a DMSO class or an intrinsic DEC value. It becomes an indexed Bellman experiment only after one fixes the latent environment, the index, the observable trajectory law, the loss, and the Bellman-error witness class whose estimation cost is being charged. Under that additional structure, the Bellman-rank state can be viewed as a compressed Bellman-sufficient representation, analogous to but distinct from a Bayesian posterior state.

### 3 Indexed information and exact AIR/MAIR identities

#### 3.1 Posterior-reference histories and index information

For a prior  $\mu$  and algorithm Alg, define the Bayesian posterior-reference trajectory law

$$\bar{\mathbb{P}}_\mu^{\text{Alg}} := \int \mathbb{P}_\omega^{\text{Alg}} \mu(d\omega).$$

A reference history is denoted

$$H'_T \sim \bar{\mathbb{P}}_\mu^{\text{Alg}}, \quad S'_t = \phi_t(H'_{t-1}, b'_t), \quad b'_t = \mathcal{L}_\mu(\Omega \mid H'_{t-1}),$$

and the algorithm's reference decision kernel is

$$p'_t = p_t(\cdot \mid H'_{t-1}).$$

**Definition 3.1** (Indexed information gain). *At state  $s$ , let  $q_s$  be the posterior law of  $Y = \chi(\Omega)$  and let  $P_{s,a}^y$  and  $P_{s,a}$  be the conditional and marginal predictive laws from Definition 2.3. For  $p \in \Delta(\mathcal{A}(s))$ , define*

$$\mathcal{I}_\chi(s, p) := \mathbb{E}_{a \sim p} \int D_{\text{KL}}(P_{s,a}^y \parallel P_{s,a}) q_s(dy). \quad (4)$$

*When  $Y = \Omega$ , this is model-index information. When  $Y = \pi^*(\Omega)$ , this is action-index information. In the action-index case, the belief is still a belief over environments; only the information target is the optimal action.*

Define the cumulative and average indexed information of an algorithm by

$$C_{\chi}^{\text{Alg}}(\mu) := \mathbb{E}_{H'_T \sim \mathbb{P}_{\mu}^{\text{Alg}}} \sum_{t=1}^T \mathcal{I}_{\chi}(S'_t, p'_t), \quad \bar{C}_{\chi}^{\text{Alg}}(\mu) := \frac{1}{T} C_{\chi}^{\text{Alg}}(\mu). \quad (5)$$

**Proposition 3.2** (Reference-history indexed chain rule). *For every prior  $\mu$ , index map  $\chi$ , and algorithm  $\text{Alg}$ ,*

$$I_{\mu}(Y; H_T) = \mathbb{E}_{H'_T \sim \mathbb{P}_{\mu}^{\text{Alg}}} \sum_{t=1}^T \mathcal{I}_{\chi}(S'_t, p'_t) = T \bar{C}_{\chi}^{\text{Alg}}(\mu). \quad (6)$$

*Proof.* The mutual-information chain rule gives

$$I_{\mu}(Y; H_T) = \sum_{t=1}^T I_{\mu}(Y; A_t, O_t \mid H_{t-1}).$$

Given  $H_{t-1}$ , the action  $A_t$  is sampled by the algorithm using only randomization independent of the environment and is conditionally independent of  $Y$ . Therefore

$$I_{\mu}(Y; A_t, O_t \mid H_{t-1}) = I_{\mu}(Y; O_t \mid H_{t-1}, A_t).$$

Conditioning further on  $A_t = a$ , the law of  $O_t$  given  $Y = y$  is  $P_{S_t, a}^y$ , while the posterior predictive law is  $P_{S_t, a}$ . Hence

$$I_{\mu}(Y; O_t \mid H_{t-1}, A_t = a) = \int D_{\text{KL}}(P_{S_t, a}^y \parallel P_{S_t, a}) q_{S_t}(dy).$$

Averaging over  $a \sim p_t(\cdot \mid H_{t-1})$  and over the marginal law of the history under the Bayesian mixture law proves the result. Writing this marginal law as the reference history law gives the displayed expression.  $\square$

### 3.2 Fixed-truth log gain and deterministic specialization

The information gain in (4) is posterior-averaged. The regret identity used in frequentist upper bounds is slightly finer: it fixes the true environment and tracks the posterior log ratio of its index.

Fix a state  $s$ , an environment  $\omega$ , and let  $y = \chi(\omega)$ . Assume domination and  $q_s(y) > 0$ . Define

$$\Lambda_{\chi}(s, a, o; y) := \log \frac{dP_{s, a}^y}{dP_{s, a}}(o).$$

For  $p \in \Delta(\mathcal{A})$ , define the fixed-truth index log gain

$$J_{\chi}(s, p; \omega) := \mathbb{E}_{a \sim p, O \sim P_{\omega, s, a}} \Lambda_{\chi}(s, a, O; \chi(\omega)). \quad (7)$$

Equivalently,

$$J_{\chi}(s, p; \omega) = \mathbb{E}_{a \sim p} \left[ D_{\text{KL}}(P_{\omega, s, a} \parallel P_{s, a}) - D_{\text{KL}}(P_{\omega, s, a} \parallel P_{s, a}^{\chi(\omega)}) \right]. \quad (8)$$

Thus the fixed-truth log gain is a pointwise-in- $\omega$  quantity, whereas  $\mathcal{I}_{\chi}(s, p)$  is a posterior-averaged index information. More explicitly, for each index value  $y$  with positive posterior mass,

$$\mathbb{E}_{\omega \sim \mu_s(\cdot \mid \chi(\omega) = y)} J_{\chi}(s, p; \omega) = \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{s, a}^y \parallel P_{s, a}),$$

and averaging this display over  $y \sim q_s$  gives  $\mathcal{I}_\chi(s, p)$ . Besides this conditional posterior averaging, a special model-index  $\chi(\omega) = \omega$  case gives  $P_{\omega, s, a} = P_{s, a}^{\chi(\omega)}$  and

$$J^{\text{MAIR}}(s, p; \omega) = \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{\omega, s, a} \| P_{s, a}),$$

In this sense,  $J_\chi(s, p; \omega)$  serves as a common fixed-truth log-gain object for stating both algorithmic information gain and fixed-truth estimation error.

For  $\eta > 0$ , define the fixed-truth indexed bracket

$$B_{\chi, \eta}(s, p; \omega) := \ell_\omega(s, p) - \frac{1}{\eta} J_\chi(s, p; \omega), \quad \ell_\omega(s, p) := \mathbb{E}_{a \sim p} \ell_\omega(s, a). \quad (9)$$

**Corollary 3.3** (Deterministic-action indexed regret identity). *Suppose the algorithm is deterministic after conditioning on  $H_{t-1}$ , so  $p_t = \delta_{a_t}$ . Fix  $\omega^*$  and  $y^* = \chi(\omega^*)$ . If the reference index posterior is updated by Bayes' rule, then conditionally on  $H_{t-1}$ ,*

$$\ell_{\omega^*}(S_t, a_t) = B_{\chi, \eta}(S_t, \delta_{a_t}; \omega^*) + \frac{1}{\eta} \mathbb{E}_{O_t \sim P_{\omega^*, S_t, a_t}} \left[ \log \frac{q_{t+1}(y^*)}{q_t(y^*)} \middle| H_{t-1} \right]. \quad (10)$$

Consequently,

$$\mathbb{E}_{\omega^*}^{\text{Alg}} L_{\omega^*}(H_T) = \frac{1}{\eta} \mathbb{E}_{\omega^*}^{\text{Alg}} \log \frac{q_{T+1}(y^*)}{q_1(y^*)} + \mathbb{E}_{\omega^*}^{\text{Alg}} \sum_{t=1}^T B_{\chi, \eta}(S_t, \delta_{a_t}; \omega^*). \quad (11)$$

The outer expectation is only over the observation process and any exogenous randomness. In a deterministic noiseless experiment the same identity is pathwise.

*Proof.* Bayes' rule gives

$$\log \frac{q_{t+1}(y^*)}{q_t(y^*)} = \log \frac{dP_{S_t, a_t}^{y^*}}{dP_{S_t, a_t}}(O_t).$$

Taking conditional expectation under the fixed truth  $\omega^*$  gives  $J_\chi(S_t, \delta_{a_t}; \omega^*)$ . Rearranging the definition of  $B_{\chi, \eta}$  proves the one-step identity. Summing over  $t$  telescopes the posterior log ratio.  $\square$

This corollary is the deterministic specialization requested in UCB analyses. It drops the outer expectation over randomized decision laws; what remains is the posterior-ratio telescope and the fixed-truth gradient bracket.

### 3.3 Exact AIR and MAIR regret identities

We now record the finite form used in the AIR/MAIR literature (Xu and Zeevi, 2025). Although we use the DMSO language to state the AIR/MAIR results and facilitate comparison with earlier work, the results in this subsection require only the Bellman-sufficient representation of Definition 2.3. There, the index may be an information index, while the state may be taken as the model posterior for DMSO; the full history remains a fallback sufficient statistic.

Let  $\mathcal{M}$  be the model class and  $\mathcal{A}$  be the decision space. Model  $M$  has reward mean  $r_M(a)$  and observation law  $P_{M, a}$ . For a candidate optimal action  $a^*$ , write

$$\Delta_{M, a^*}(p) := r_M(a^*) - \mathbb{E}_{a \sim p} r_M(a).$$

**AIR.** An AIR belief is a pair belief  $\nu \in \Delta(\mathcal{M} \times \mathcal{A})$  over  $(M, A^*)$ , while the reference is an action marginal  $q \in \text{int } \Delta(\mathcal{A})$ . Define

$$\text{AIR}_{q,\eta}(p, \nu) := \mathbb{E}_{(M, A^*) \sim \nu} \Delta_{M, A^*}(p) - \frac{1}{\eta} \mathbb{E}_{a \sim p} I_\nu(A^*; O | a) - \frac{1}{\eta} D_{\text{KL}}(q_\nu \| q), \quad (12)$$

where  $q_\nu$  is the  $A^*$ -marginal of  $\nu$ . Equivalently,

$$\text{AIR}_{q,\eta}(p, \nu) = \mathbb{E}_{(M, A^*), a, O} \left[ \Delta_{M, A^*}(\delta_a) - \frac{1}{\eta} \log \frac{\nu(A^* | a, O)}{q(A^*)} \right].$$

**Lemma 3.4** (AIR gradient bracket). *Assume finite spaces, or dominated spaces where the displayed Gateaux derivatives exist. Up to an additive constant on the simplex,*

$$\frac{\partial}{\partial \nu(M, a^*)} \text{AIR}_{q,\eta}(p, \nu) = \Delta_{M, a^*}(p) - \frac{1}{\eta} \mathbb{E}_{a \sim p, O \sim P_{M,a}} \log \frac{\nu(A^* = a^* | a, O)}{q(a^*)}. \quad (13)$$

Consequently, for every fixed pair  $(M, a^*)$ ,

$$B_{q,\eta}^{\text{AIR}}(p, \nu; M, a^*) := \text{AIR}_{q,\eta}(p, \nu) + \langle \nabla_\nu \text{AIR}_{q,\eta}(p, \nu), \delta_{(M, a^*)} - \nu \rangle \quad (14)$$

$$= \Delta_{M, a^*}(p) - \frac{1}{\eta} \mathbb{E}_{a \sim p, O \sim P_{M,a}} \log \frac{\nu(A^* = a^* | a, O)}{q(a^*)}. \quad (15)$$

*Proof.* For fixed  $a$ , set

$$w_{a^*, o} = \sum_M \nu(M, a^*) P_{M,a}(o), \quad w_o = \sum_{a^*} w_{a^*, o}.$$

Then

$$I_\nu(A^*; O | a) + D_{\text{KL}}(q_\nu \| q) = \sum_{a^*, o} w_{a^*, o} \log \frac{w_{a^*, o}}{w_o q(a^*)}.$$

Differentiating with respect to  $\nu(M, a^*)$  gives

$$\mathbb{E}_{O \sim P_{M,a}} \log \frac{\nu(A^* = a^* | a, O)}{q(a^*)},$$

because the  $+1$  terms from differentiating  $w_{a^*, o} \log w_{a^*, o}$  and  $w_o \log w_o$  cancel. Adding the derivative of the linear regret term proves (13). Pairing the gradient with  $\delta_{(M, a^*)} - \nu$  cancels the posterior-averaged terms and gives (14).  $\square$

**Theorem 3.5** (Exact AIR regret identity). *Let an algorithm generate decision laws  $p_t$ , pair beliefs  $\nu_t$ , and reference action beliefs  $q_t \in \text{int } \Delta(\mathcal{A})$ . After action  $a_t$  and observation  $O_t$ , set*

$$q_{t+1}(\cdot) = \nu_t(A^* \in \cdot | a_t, O_t).$$

*Fix a model  $M^*$  and a comparator  $a^*$  with  $q_t(a^*) > 0$  for all relevant  $t$ ; in the standard stochastic case, take  $a^* \in \arg \max_a r_{M^*}(a)$ . Let*

$$\text{Reg}_T(\text{Alg}, M^*, a^*) := \mathbb{E} \sum_{t=1}^T (r_{M^*}(a^*) - r_{M^*}(A_t)).$$

*Then, for every  $\eta > 0$ ,*

$$\text{Reg}_T(\text{Alg}, M^*, a^*) = \frac{1}{\eta} \mathbb{E} \log \frac{q_{T+1}(a^*)}{q_1(a^*)} + \mathbb{E} \sum_{t=1}^T B_{q_t, \eta}^{\text{AIR}}(p_t, \nu_t; M^*, a^*). \quad (16)$$

**Remark 3.6** (Adversarial model sequence). *Since Lemma 3.4 holds pointwise for every model–action pair, and does not require the true model to be stochastic, Theorem 3.5 extends directly to an adversarial model sequence  $\mathbf{M} = (M_1, \dots, M_T) \in \mathcal{M}^T$ . In this case, the comparator is the best-in-hindsight action  $a^*$ , and the regret identity remains exact. We omit this extension only to avoid additional notation.*

*Proof.* Condition on  $H_{t-1}$ . By the definition of the action-marginal posterior,

$$\mathbb{E} \left[ \log \frac{q_{t+1}(a^*)}{q_t(a^*)} \middle| H_{t-1} \right] = \mathbb{E}_{a \sim p_t, O \sim P_{M^*, a}} \log \frac{\nu_t(A^* = a^* \mid a, O)}{q_t(a^*)}.$$

Combining this equality with (14) gives a one-step equality between regret, bracket, and posterior-ratio increment. Taking expectations and summing over time telescopes the logarithm.  $\square$

**MAIR.** For MAIR the belief is  $\mu \in \Delta(\mathcal{M})$  and the reference is  $\rho \in \text{int } \Delta(\mathcal{M})$ . Write  $\mu_{O|a} = \int P_{M,a} \mu(dM)$  and

$$\Delta_M(p) = r_M(a_M^*) - \mathbb{E}_{a \sim p} r_M(a), \quad a_M^* \in \arg \max_a r_M(a).$$

Define

$$\text{MAIR}_{\rho, \eta}(p, \mu) := \mathbb{E}_{M \sim \mu} \Delta_M(p) - \frac{1}{\eta} \mathbb{E}_{a \sim p} I_\mu(M; O \mid a) - \frac{1}{\eta} D_{\text{KL}}(\mu \parallel \rho). \quad (17)$$

The same differentiation gives

$$B_{\rho, \eta}^{\text{MAIR}}(p, \mu; M^*) := \text{MAIR}_{\rho, \eta}(p, \mu) + \langle \nabla_\mu \text{MAIR}_{\rho, \eta}(p, \mu), \delta_{M^*} - \mu \rangle \quad (18)$$

$$= \Delta_{M^*}(p) - \frac{1}{\eta} \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{M^*, a} \parallel \mu_{O|a}) - \frac{1}{\eta} \log \frac{\mu(M^*)}{\rho(M^*)}. \quad (19)$$

If  $\rho_{t+1}(\cdot) = \mu_t(\cdot \mid a_t, O_t)$ , then

$$\text{Reg}_T(\text{Alg}, M^*) = \frac{1}{\eta} \mathbb{E} \log \frac{\rho_{T+1}(M^*)}{\rho_1(M^*)} + \mathbb{E} \sum_{t=1}^T B_{\rho_t, \eta}^{\text{MAIR}}(p_t, \mu_t; M^*). \quad (20)$$

The proof is the same posterior-ratio telescope as for AIR. Thus AIR and MAIR are one indexed identity with different choices of  $Y$ .

**Remark 3.7** (Danskin, Nash equilibrium, and why the bracket is not accidental). *The direct proofs above are intentionally short. The original AIR proof of Xu and Zeevi (2025, Section 5) gives the variational reason behind the same equations. Introduce an auxiliary prediction rule  $Q : (a, o) \mapsto \Delta(\mathcal{A})$  and the posterior-scoring objective*

$$\mathbb{E}_{(M, A^*), a, O} \left[ \Delta_{M, A^*}(\delta_a) - \frac{1}{\eta} \log \frac{Q[a, O](A^*)}{q(A^*)} \right].$$

*For fixed  $\nu$ , the logarithmic score is strongly convex in the predictive distribution and is uniquely minimized by the Bayesian posterior  $Q_\nu[a, o] = \nu(A^* \in \cdot \mid a, o)$ . The optimized value is the AIR objective; the unoptimized objective is linear, hence concave, in  $\nu$ . Danskin’s theorem identifies the AIR gradient with the derivative at the optimizer  $Q_\nu$ . This is the conceptual content of Lemma 5.3 in Xu and Zeevi (2025). The same concavity plus strong convexity also gives the Nash equilibrium statement of Lemma 5.2 by first-order/Danskin reasoning, without invoking a separate fixed-point theorem. This does not necessarily simplify the full proof, because compactness, differentiability, and boundary conditions still have to be checked. It does explain why the gradient bracket is not an accidental cancellation: posterior scoring produces the minimizer, Danskin produces the gradient, and the regret identity telescopes the posterior reference.*

The perspective developed in [Xu and Zeevi \(2025\)](#) is that frequentist sequential learning can be organized around algorithmic beliefs  $\mu$ , interpreted as predictive laws, and decision rules induced by reference or posterior beliefs  $\rho$ , either through posterior sampling or through robust minimax optimization. We use MAIR to illustrate how a single functional captures both the confidence multiplier in UCB-type algorithms and the decision-estimation coefficient in one-round minimax rules.

**Lemma 3.8** (MAIR gradient bracket and three canonical specializations). *Assume the observation laws are dominated and the displayed derivatives exist. For fixed  $p, \rho, \mu$ , up to an additive constant on the probability simplex,*

$$\frac{\partial}{\partial \mu(M)} \text{MAIR}_{\rho, \eta}(p, \mu) = \Delta_M(p) - \frac{1}{\eta} \log \frac{\mu(M)}{\rho(M)} - \frac{1}{\eta} \mathbb{E}_{\pi \sim p} D_{\text{KL}}(P_{M, \pi} \| \mu_{o|\pi}).$$

Consequently, for every  $M^*$ ,

$$\begin{aligned} & \text{MAIR}_{\rho, \eta}(p, \mu) + \langle \nabla_{\mu} \text{MAIR}_{\rho, \eta}(p, \mu), \delta_{M^*} - \mu \rangle \\ &= \Delta_{M^*}(p) - \frac{1}{\eta} \mathbb{E}_{\pi \sim p} D_{\text{KL}}(P_{M^*, \pi} \| \mu_{o|\pi}) - \frac{1}{\eta} \log \frac{\mu(M^*)}{\rho(M^*)}. \end{aligned} \quad (21)$$

This bracket has three important specializations.

1. If  $\mu = \rho$ , the logarithmic correction vanishes and the bracket becomes the KL-DEC offset

$$\Delta_{M^*}(p) - \frac{1}{\eta} \mathbb{E}_{\pi \sim p} D_{\text{KL}}(P_{M^*, \pi} \| \mu_{o|\pi}). \quad (22)$$

In [Section 4](#) we will show that for GP-UCB,

$$\text{self-normalized calibration} + \text{UCB optimism} \Rightarrow \text{MAIR gradient-error condition.}$$

2. If  $\bar{\mu} \in \arg \max_{\mu \in \Delta(\mathcal{M})} \text{MAIR}_{\rho, \eta}(p, \mu)$  is an interior maximizer, then

$$\text{MAIR}_{\rho, \eta}(p, \bar{\mu}) + \langle \nabla_{\mu} \text{MAIR}_{\rho, \eta}(p, \bar{\mu}), \delta_{M^*} - \bar{\mu} \rangle \leq \text{MAIR}_{\rho, \eta}(p, \bar{\mu}). \quad (23)$$

This is the MAMS/MEBO bracket control.

3. Let  $p_{M, \pi} = dP_{M, \pi} / d\nu_{\pi}$  and define, for an auxiliary pair  $(\bar{\pi}, \bar{o})$ ,

$$\frac{d\mu_{\bar{\pi}, \bar{o}}^{1/2}}{d\rho}(M) := \frac{\sqrt{p_{M, \bar{\pi}}(\bar{o})}}{\int \sqrt{p_{N, \bar{\pi}}(\bar{o})} \rho(dN)}. \quad (24)$$

Let  $h^2(P, Q) := 1 - \int \sqrt{dP dQ}$ . If  $\bar{\pi} \sim p$  and  $\bar{o} \sim P_{M^*, \bar{\pi}}$ , then

$$\begin{aligned} & \mathbb{E}_{\bar{\pi}, \bar{o}} \left[ \text{MAIR}_{\rho, \eta}(p, \mu_{\bar{\pi}, \bar{o}}^{1/2}) + \langle \nabla_{\mu} \text{MAIR}_{\rho, \eta}(p, \mu_{\bar{\pi}, \bar{o}}^{1/2}), \delta_{M^*} - \mu_{\bar{\pi}, \bar{o}}^{1/2} \rangle \right] \\ & \leq \Delta_{M^*}(p) - \frac{1}{\eta} \mathbb{E}_{\pi \sim p} \mathbb{E}_{M \sim \rho} h^2(P_{M^*, \pi}, P_{M, \pi}). \end{aligned} \quad (25)$$

Thus the square-root posterior turns the MAIR bracket into a Hellinger E2D/DEC offset.

*Proof.* The derivative calculation is direct. The derivative of  $\mathbb{E}_\mu \Delta_M(p)$  is  $\Delta_M(p)$ ; the derivative of  $D_{\text{KL}}(\mu \parallel \rho)$  is  $\log(\mu(M)/\rho(M)) + 1$ ; and differentiating the predictive mixture in the mutual-information term gives  $D_{\text{KL}}(P_{M,\pi} \parallel \mu_{o|\pi})$ , up to the same simplex constant. Pairing the derivative with  $\delta_{M^*} - \mu$  cancels the averaged terms in (17) and proves (21).

The case  $\mu = \rho$  is immediate from the logarithmic cancellation.

The case  $\bar{\mu} \in \arg \max \text{MAIR}$  follows from first-order optimality.

It remains to justify the square-root claim; this is a standard fact for the squared Hellinger distance. For fixed  $\bar{\pi}$ , let

$$Z_{\bar{\pi}}(\bar{o}) := \int \sqrt{p_{N,\bar{\pi}}(\bar{o})} \rho(dN).$$

Then

$$\log \frac{d\mu_{\bar{\pi},\bar{o}}^{1/2}}{d\rho}(M^*) = \frac{1}{2} \log p_{M^*,\bar{\pi}}(\bar{o}) - \log Z_{\bar{\pi}}(\bar{o}).$$

Under  $\bar{o} \sim P_{M^*,\bar{\pi}}$ , Jensen's inequality gives

$$\begin{aligned} \mathbb{E} \log \frac{d\mu_{\bar{\pi},\bar{o}}^{1/2}}{d\rho}(M^*) &= -\mathbb{E} \log \frac{Z_{\bar{\pi}}(\bar{o})}{\sqrt{p_{M^*,\bar{\pi}}(\bar{o})}} \\ &\geq -\log \int \sqrt{p_{M^*,\bar{\pi}}(o)} Z_{\bar{\pi}}(o) d\nu_{\bar{\pi}}(o) \\ &= -\log (1 - \mathbb{E}_{M \sim \rho} h^2(P_{M^*,\bar{\pi}}, P_{M,\bar{\pi}})) \\ &\geq \mathbb{E}_{M \sim \rho} h^2(P_{M^*,\bar{\pi}}, P_{M,\bar{\pi}}). \end{aligned}$$

Averaging over  $\bar{\pi} \sim p$ , substituting into (21), and dropping the nonnegative KL term proves (25).  $\square$

## 4 Upper bounds: one identity and four algorithm families

We first isolate the fixed-truth log-potential telescope, because it is the pathwise algebra underneath AIR/MAIR upper bounds. Bayesian mutual information, UCB self-normalized gain, and EBO log-partition relaxations are different ways of controlling this telescope; they should not be conflated. Fix a truth  $\omega^*$ , write  $y^* = \chi(\omega^*)$ , and let  $q_s$  be the maintained reference law of the index at state  $s$ . This reference law may be an exact posterior marginal, an algorithmic belief marginal, a confidence-to-belief construction, or an exponential-weights marginal. After action  $a$  and observation  $o$ , write  $q^+(\cdot \mid s, a, o)$  for the next reference marginal. The fixed-truth reference log gain is

$$\mathbf{G}_\chi(s, p; \omega^*) := \mathbb{E}_{a \sim p, O \sim P_{\omega^*, s, a}} \log \frac{q^+(y^* \mid s, a, O)}{q_s(y^*)}. \quad (26)$$

When the reference update is exact indexed Bayes, this is  $J_\chi(s, p; \omega^*)$  from (7). When the reference is an AIR pair belief  $\nu$  with marginal  $q$ , it is the AIR posterior-coordinate log ratio

$$\mathbb{E}_{a \sim p, O \sim P_{\omega^*, a}} \log \frac{\nu(Y = y^* \mid a, O)}{q(y^*)}.$$

For any predictable continuation potential  $V_t$  on states and any coefficient  $\gamma > 0$ , define the specialized fixed-truth logarithmic potential

$$\Phi_t^{V,\gamma}(s; y^*) := V_t(s) + \gamma \log \frac{1}{q_s(y^*)}. \quad (27)$$

The corresponding AIR/MAIR Bellman bracket is

$$\mathfrak{B}_\chi^{V,\gamma}(s, p; \omega^\star) := \ell_{\omega^\star}(s, p) + \mathbb{E}_{\omega^\star}[V_{t+1}(S^+) \mid s, p] - V_t(s) - \gamma \mathbf{G}_\chi(s, p; \omega^\star), \quad (28)$$

where  $S^+ = \tau(s, A, O)$  with  $A \sim p$  and  $O \sim P_{\omega^\star, s, A}$ . The identity behind the whole upper section is the telescope

$$\begin{aligned} \mathbb{E}_{\omega^\star} L_{\omega^\star}(H_T) &= V_1(s_1) - \mathbb{E}_{\omega^\star} V_{T+1}(S_{T+1}) + \gamma \log \frac{1}{q_1(y^\star)} - \gamma \mathbb{E}_{\omega^\star} \log \frac{1}{q_{T+1}(y^\star)} \\ &\quad + \mathbb{E}_{\omega^\star} \sum_{t=1}^T \mathfrak{B}_\chi^{V,\gamma}(S_t, p_t; \omega^\star). \end{aligned} \quad (29)$$

Thus an upper bound is a Bellman supersolution for this log-potential bracket. The four algorithm families below differ only in how they make the bracket small: the exact log-penalized Bellman program optimizes it dynamically; UCB certifies it by calibration and optimism; E2D/DEC drops or bounds the continuation to obtain a one-step offset; and AMS/EBO optimizes a robust KL-dual belief relaxation whose first-order condition certifies it.

**Theorem 4.1** (Generic fixed-truth indexed AIR/MAIR upper certificate). *Fix  $\omega^\star$  and  $y^\star = \chi(\omega^\star)$ . Assume  $q_1(y^\star) > 0$  and that the reference update remains positive on  $y^\star$  along the analyzed trajectory. If an algorithm chooses  $p_t$  so that, for every reached state,*

$$\mathfrak{B}_\chi^{V,\gamma}(S_t, p_t; \omega^\star) \leq \varepsilon_t,$$

then

$$\mathbb{E}_{\omega^\star}^{\text{Alg}} L_{\omega^\star}(H_T) \leq V_1(s_1) + \gamma \log \frac{1}{q_1(y^\star)} - \mathbb{E}_{\omega^\star}^{\text{Alg}} V_{T+1}(S_{T+1}) - \gamma \mathbb{E}_{\omega^\star}^{\text{Alg}} \log \frac{1}{q_{T+1}(y^\star)} + \mathbb{E}_{\omega^\star}^{\text{Alg}} \sum_{t=1}^T \varepsilon_t. \quad (30)$$

*Proof.* Taking the conditional expectation of  $\gamma \log(1/q_{t+1}(y^\star)) - \gamma \log(1/q_t(y^\star))$  under  $P_{\omega^\star, s, a}$  gives  $-\gamma \mathbf{G}_\chi(s, p; \omega^\star)$ . Substituting the bracket inequality into the telescoping identity (29) proves (30).  $\square$

It is useful to keep one concrete state specialization in mind. A fixed estimation procedure maps histories to algorithmic beliefs  $\nu_t$  over environments, a decision procedure maps the retained state to  $p_t$ , and the index marginal is

$$q_t = \chi_\# \nu_t. \quad (31)$$

In a well-specified Bayes analysis,  $\nu_t$  is the exact posterior. In a frequentist analysis,  $\nu_t$  may be only an algorithmic belief or confidence object; then a calibration event must put the fixed truth into the comparison set used by the robust Bellman program below. Posterior averaging of the coordinate potential gives

$$\mathbb{E}_{Y \sim \nu_\chi} \gamma \log \frac{1}{q_s(Y)} = \gamma H(\nu_\chi) + \gamma D_{\text{KL}}(\nu_\chi \| q_s), \quad (32)$$

and for a finite index set the same logarithmic scoring rule has the KL-dual log-partition form

$$\Psi_\gamma(q, z) := \gamma \log \sum_{y \in \mathcal{Y}} q(y) e^{zy/\gamma} = \sup_{r \in \Delta(\mathcal{Y})} \{ \langle r, z \rangle - \gamma D_{\text{KL}}(r \| q) \}. \quad (33)$$



The coordinate log loss, its posterior average, and the dual log partition are three representations of the same KL/logarithmic scoring rule. They are not interchangeable in a proof: fixed-truth guarantees use (27), posterior-averaged Bayes identities use (32), and AMS/EBO relaxations use (33).

In the model-index specialization, write the posterior/reference belief as  $\mu_t$ . If the maintained reference update is exact Bayes, then  $G_\chi = J_\chi$  and the fixed-truth MAIR information increment is

$$J_t^{\text{MAIR}}(p; \omega^*) := \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{\omega^*, t}(\cdot | S_t, a) \| P_{\mu_t, t}(\cdot | S_t, a)). \quad (34)$$

The action-index AIR version is identical with the index  $Y = A^*$ , the reference marginal  $q_t$ , and the AIR posterior coordinate  $q^+(a^* | a, o)$  in place of the model posterior.

#### 4.1 Information-potential Bellman programming.

The exact frequentist upper algorithm is a log-penalized Bellman program on calibrated state/index representations. Let  $\mathfrak{C}_t(s)$  be the set of environments that the algorithmic state regards as admissible at state  $s$ . The set may be a confidence set, a localized posterior support, or a robust belief support. For  $\gamma > 0$ , define

$$W_{T+1}^\gamma(s) = 0, \quad W_t^\gamma(s) := \inf_{p \in \Delta(\mathcal{A}(s))} \sup_{\omega \in \mathfrak{C}_t(s)} \{ \ell_\omega(s, p) + \mathbb{E}_\omega[W_{t+1}^\gamma(S^+) | s, p] - \gamma G_\chi(s, p; \omega) \}. \quad (35)$$

A policy that attains the infimum in (35) is the information-penalized Bellman dynamic-programming algorithm. It is the direct interactive analogue of choosing an information-risk or Gibbs estimator in a fixed experiment: the logarithmic mass term is now propagated through a controlled Bellman recursion.

**Theorem 4.2** (Frequentist log-penalized Bellman upper bound). *Fix  $\Omega_0 \subseteq \Omega$  and  $\gamma > 0$ . Suppose the reference update and the comparison sets  $\mathfrak{C}_t$  are calibrated for  $\Omega_0$  in the following sense: for every  $\omega^* \in \Omega_0$ , on the calibration event  $\mathcal{E}$ ,  $\omega^* \in \mathfrak{C}_t(S_t)$  whenever round  $t$  is reached. Suppose the algorithm chooses  $p_t$  so that*

$$\sup_{\omega \in \mathfrak{C}_t(S_t)} \{ \ell_\omega(S_t, p_t) + \mathbb{E}_\omega[W_{t+1}^\gamma(S^+) | S_t, p_t] - \gamma G_\chi(S_t, p_t; \omega) \} \leq W_t^\gamma(S_t) + \varepsilon_t.$$

*Then, for every fixed truth  $\omega^* \in \Omega_0$  with  $y^* = \chi(\omega^*)$  and  $q_1(y^*) > 0$ ,*

$$\mathbb{E}_{\omega^*}^{\text{Alg}} L_{\omega^*}(H_T) \leq W_1^\gamma(s_1) + \gamma \log \frac{1}{q_1(y^*)} + \mathbb{E}_{\omega^*}^{\text{Alg}} \sum_{t=1}^T \varepsilon_t + \text{cal}_T(\omega^*), \quad (36)$$

*provided the index is finite or countable, where*

$$\text{cal}_T(\omega^*) := \mathbb{E}_{\omega^*}[L_{\omega^*}(H_T) \mathbb{1}\{\mathcal{E}^c\}].$$

*If calibration holds surely, then  $\text{cal}_T(\omega^*) = 0$ ; if losses are bounded by  $L$ , then  $\text{cal}_T(\omega^*) \leq LT\mathbb{P}_{\omega^*}(\mathcal{E}^c)$ .*

*Proof.* On the calibration event, the fixed truth is one of the environments in the robust supremum. Therefore the displayed approximate minimization implies

$$\ell_{\omega^*}(S_t, p_t) + \mathbb{E}_{\omega^*}[W_{t+1}^\gamma(S^+) | S_t, p_t] - W_t^\gamma(S_t) - \gamma G_\chi(S_t, p_t; \omega^*) \leq \varepsilon_t.$$

Apply Theorem 4.1 with  $V_t = W_t^\gamma$ . Since  $W_{T+1}^\gamma = 0$  and the terminal index log loss is nonnegative for a finite or countable index, the result follows on  $\mathcal{E}$ . Splitting the expectation according to  $\mathcal{E}$  and  $\mathcal{E}^c$  gives the calibration term.  $\square$

The posterior-averaged Bayesian recursion is a specialization rather than the primitive fixed-truth statement. If the robust comparison set is replaced by the current posterior law and one averages the coordinate identity over  $Y \sim q_s$ , then

$$\mathbb{E}[H_\chi(S_{t+1}) \mid s, p] - H_\chi(s) = -\mathcal{I}_\chi(s, p).$$

For a fixed multiplier  $\gamma > 0$ , define the Bayes information-regularized value

$$U_{T+1}^\gamma \equiv 0, \quad U_t^\gamma(s) := \inf_{p \in \Delta(\mathcal{A}(s))} \{ \ell(s, p) + \mathbb{E}[U_{t+1}^\gamma(S_{t+1}) \mid s, p] - \gamma \mathcal{I}_\chi(s, p) \}. \quad (37)$$

Equivalently,

$$\Phi_t^\gamma(s) := U_t^\gamma(s) + \gamma H_\chi(s) \quad (38)$$

solves the admissible-relaxation recursion

$$\Phi_t^\gamma(s) = \inf_{p \in \Delta(\mathcal{A}(s))} \{ \ell(s, p) + \mathbb{E}[\Phi_{t+1}^\gamma(S_{t+1}) \mid s, p] \}, \quad \Phi_{T+1}^\gamma(s) = \gamma H_\chi(s). \quad (39)$$

**Theorem 4.3** (Posterior-averaged information-potential upper bound). *Assume the indexed Bellman representation is exact under the Bayesian mixture law and  $H_\mu(Y) < \infty$ . If a policy chooses  $p_t$  satisfying*

$$\ell(S_t, p_t) + \mathbb{E}[U_{t+1}^\gamma(S_{t+1}) \mid S_t, p_t] - \gamma \mathcal{I}_\chi(S_t, p_t) \leq U_t^\gamma(S_t) + \varepsilon_t,$$

then

$$\mathbb{E}L_\Omega(H_T) \leq U_1^\gamma(s_1) + \gamma I_\mu(Y; H_T) + \sum_{t=1}^T \varepsilon_t \leq U_1^\gamma(s_1) + \gamma H_\mu(Y) + \sum_{t=1}^T \varepsilon_t. \quad (40)$$

*This is the posterior average of the fixed-truth coordinate telescope, not a pointwise replacement for Theorem 4.2.*

*Proof.* Rearrange the displayed inequality and telescope  $U_t^\gamma$ . Proposition 3.2 identifies  $\mathbb{E} \sum_t \mathcal{I}_\chi(S_t, p_t)$  with  $I_\mu(Y; H_T)$ , and  $I_\mu(Y; H_T) \leq H_\mu(Y)$  gives the final bound. The equivalence with (39) follows from the posterior entropy-drop identity.  $\square$

**Theorem 4.4** (Frequentist Bellman information-risk matching principle). *Fix a problem class  $\Omega_0$ , a horizon  $T$ , an index  $\chi$ , and a Bellman-sufficient state/reference update. Suppose the upper side has a calibrated log-penalized Bellman program as in Theorem 4.2. Define its worst-case upper value*

$$\mathfrak{U}_T(\Omega_0) := \inf_{\gamma > 0, q_1, \mathfrak{C}, \text{Alg}} \sup_{\omega \in \Omega_0} \left\{ W_1^\gamma(s_1) + \gamma \log \frac{1}{q_1(\chi(\omega))} + \text{cal}_T(\omega) + \text{opt}_T(\omega) \right\},$$

*where the infimum is over initial reference laws, calibrated comparison mechanisms, and algorithms that approximately solve the displayed Bellman recursion, and where*

$$\text{opt}_T(\omega) := \mathbb{E}_\omega^{\text{Alg}} \sum_{t=1}^T \varepsilon_t$$

*records optimization or relaxation error. For the exact log-penalized Bellman dynamic program,  $\text{opt}_T \equiv 0$ . Then*

$$\mathfrak{R}_T^*(\Omega_0) \leq \mathfrak{U}_T(\Omega_0).$$

Conversely, suppose there are a prior  $\mu$  supported on  $\Omega_0$ , the same index  $\chi$ , the same state process, and Bellman-Fano lower certificates  $\overline{C}, \overline{\Gamma}^r$  satisfying the critical condition (60). Then

$$\mathfrak{R}_T^*(\Omega_0) \geq \frac{Tr}{2}.$$

Consequently, if for the same representation and some radius  $r$  one has

$$\mathfrak{U}_T(\Omega_0) \leq C_{\text{up}} Tr \quad \text{and} \quad \overline{C}_1(s_1) + \log 2 \leq \frac{1}{2} \mathcal{E}_1^r(\mu, \chi),$$

then the frequentist minimax regret is sandwiched as

$$\frac{Tr}{2} \leq \mathfrak{R}_T^*(\Omega_0) \leq C_{\text{up}} Tr.$$

This is the precise sense in which the log-penalized Bellman dynamic program matches the Bellman-Fano lower bound. The theorem is conditional on representational sufficiency and calibration; it is not a claim that an arbitrary index, belief, or one-step relaxation is sharp.

*Proof.* The upper inequality is Theorem 4.2 followed by taking the supremum over  $\omega \in \Omega_0$  and then optimizing over the displayed design choices. The lower inequality is Theorem 5.3 and Yao's principle, since  $\mu$  is supported on  $\Omega_0$ . The sandwich conclusion is the conjunction of the two bounds. The ghost history enters only through the lower certificate; the upper side may use the exact Bellman recursion or any admissible relaxation, including random-payout or sequential symmetrization, that upper bounds  $W_1^\gamma$ .  $\square$

Theorem 4.4 is the interactive analogue of noninteractive information-risk minimization. In a fixed experiment, the statistical radius is determined by balancing a loss scale against a log prior mass or metric entropy. Here the loss scale is propagated by Bellman recursion, the upper log mass is the coordinate cost  $\log(1/q_1(\chi(\omega)))$ , and the lower entropy is the ghost-good quantile  $-\log p_r$ . The exact recursion is often computationally difficult, but it is the benchmark that the following algorithm families approximate or certify.

## 4.2 UCB families: calibration plus optimism.

An upper-confidence-bound (UCB) algorithm maintains a prediction and an uncertainty scale, builds a confidence band from them, and chooses an action that is optimistic within that band (Auer et al., 2002; Abbasi-Yadkori et al., 2011; Srinivas et al., 2010). We use the following notation throughout. The unmultiplied uncertainty scale is denoted by  $\sigma_t(a)$  in the generic discussion and by  $s_t(x)$  in the GP specialization below. The confidence half-width is

$$w_t(a) = \beta_t \sigma_t(a),$$

where  $\beta_t$  is the confidence multiplier. Calibration says that the fixed truth is contained in the band of radius  $w_t(a)$ ; optimism says that the selected action maximizes the upper envelope. Together, in reward-maximization problems, these two facts give

$$r_{\omega^*}(s, a^*) \leq m_t(a^*) + w_t(a^*) \leq m_t(a_t) + w_t(a_t) \leq r_{\omega^*}(s, a_t) + 2w_t(a_t),$$

and hence the instantaneous regret is at most  $2\beta_t \sigma_t(a_t)$ . An elliptical-potential/log-determinant lemma then controls the realized sum of  $\sigma_t^2(a_t)$ . Thus UCB does not solve the AIR/MAIR bracket optimization directly; it certifies the relevant fixed-truth bracket after the optimistic action has been chosen.

**Lemma 4.5** (Calibration and optimism imply bracket control). *Fix a truth  $\omega^*$  and condition on an event  $\mathcal{E}$  on which the algorithm’s state is calibrated for that truth. Suppose that, for every reached round  $t$ , the action  $a_t$  chosen by the algorithm and a predictable uncertainty scale  $\sigma_t(a) \geq 0$  satisfy*

$$\ell_{\omega^*}(S_t, a_t) \leq c_{\text{opt}} \beta_t \sigma_t(a_t), \quad (41)$$

where  $c_{\text{opt}}$  is a numerical optimism constant; in the usual two-sided reward-confidence proof above,  $c_{\text{opt}} = 2$ . Then for every  $\eta > 0$ ,

$$\ell_{\omega^*}(S_t, a_t) - \frac{1}{\eta} \sigma_t^2(a_t) \leq \frac{\eta c_{\text{opt}}^2 \beta_t^2}{4}. \quad (42)$$

Consequently, for the deterministic-action fixed-truth bracket in (9),

$$B_{\chi, \eta}(S_t, \delta_{a_t}; \omega^*) \leq \frac{\eta c_{\text{opt}}^2 \beta_t^2}{4} + \frac{1}{\eta} (\sigma_t^2(a_t) - J_{\chi}(S_t, \delta_{a_t}; \omega^*)), \quad (43)$$

where the fixed-truth index log gain is defined in (7). If, in addition, the log-determinant/elliptical-potential argument gives a realized bound

$$\sum_{t=1}^T \sigma_t^2(a_t) \leq c_{\text{sn}} G_T,$$

for an information telescope  $G_T$ , then summing (43) reduces UCB regret control to the comparison between  $G_T$  and the fixed-truth log-gain telescope.

*Proof.* The optimism/calibration hypothesis (41) and Young’s inequality give

$$c_{\text{opt}} \beta_t \sigma_t(a_t) \leq \frac{\eta c_{\text{opt}}^2 \beta_t^2}{4} + \frac{\sigma_t^2(a_t)}{\eta}.$$

This proves (42). Subtracting  $\eta^{-1} J_{\chi}(S_t, \delta_{a_t}; \omega^*)$  from both sides and using the definition of  $B_{\chi, \eta}$  proves (43). The final sentence is the result of summing the displayed inequality over time.  $\square$

The correction  $\sigma_t^2(a_t) - J_{\chi}(S_t, \delta_{a_t}; \omega^*)$  is not a second estimation cost. It is the algebraic difference between the tractable self-normalized variance proxy and the fixed-truth posterior log gain appearing in the bracket. In linear-UCB and GP-UCB proofs, the log-determinant or elliptical-potential argument is precisely the certificate that the realized sum of  $\sigma_t^2(a_t)$  is controlled by an information telescope. The confidence multiplier  $\beta_t$  is separate: it is the price of converting calibrated uncertainty into regret through optimism. Therefore

$$\text{self-normalized calibration} + \text{optimism} \implies \text{fixed-truth AIR/MAIR bracket control}.$$

In this organizing perspective, the round-heterogeneous summation of unmultiplied uncertainty scales, usually controlled by an elliptical-potential or log-determinant argument, plays the role of a realized estimation-complexity telescope. The confidence multiplier enters instead through the regret-to-uncertainty conversion, and hence through the coefficient in the fixed-truth bracket. This separates two mechanisms that are often blurred in informal UCB discussions: posterior or confidence widths quantify the “estimation complexity” of the underlying parameter, as explained by Lattimore (2023), while optimism and calibration determine how that geometry pays for regret. The resulting unification reveals a common posterior-telescoping structure while keeping distinct the estimation telescope and the coefficient mechanisms behind UCB, E2D, and AMS/EBO.

### 4.3 E2D/DEC: robust one-step offset optimization.

The decision-estimation coefficient (DEC) (Foster et al., 2021) and estimation-to-decision (E2D) algorithms choose  $p$  by solving a one-round robust offset. In model-index notation the schematic KL form is

$$\inf_{p \in \Delta(\mathcal{A})} \sup_{\omega \in \mathcal{N}(s)} \{ \ell_\omega(s, p) - \gamma \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{\omega, s, a} \| P_{\rho, s, a}) \}, \quad (44)$$

where  $\rho$  is a reference belief or reference model and  $\mathcal{N}(s)$  is a localized comparison set that contains the fixed truth on the calibration event. The same one-round minimax principle applies to the original Hellinger form and constrained variants (Foster et al., 2023).

What E2D/DEC keeps is the immediate regret and a one-step statistical-separation penalty. What it drops, freezes, or upper bounds is the Bellman continuation term

$$\mathbb{E}_\omega[V_{t+1}(S^+) \mid s, p] - V_t(s),$$

together with the exact coordinate reference log gain  $G_\chi$  in (28). Thus E2D/DEC is not an optimizer of the full Bellman potential unless one proves that the omitted continuation is dominated by the displayed one-step penalty. In an exact Bayesian calculation, posterior averaging can turn the KL penalty into a one-step mutual-information term. In a fixed-truth or frequentist calculation, the corresponding statement is the reference posterior-ratio identity for  $G_\chi$ , or its exact-Bayes specialization  $J_\chi$ , plus whatever calibration or localization is needed to compare  $P_{\rho, s, a}$  with the reference predictive law in that identity. Without this compatibility, one may pay for estimation difficulty once through the one-step DEC and again through a separate log-determinant or posterior-ratio telescope. The safe interpretation is therefore: DEC/E2D is a one-step relaxation, re-solved at each state, of the fixed-truth log-posterior Bellman bracket.

### 4.4 AMS/EBO: robust convex belief optimization.

AMS/EBO should be read as a tractable relaxation of the robust log-penalized Bellman program. Work with an action or policy index  $Y = \chi(\Omega)$ . At state  $s$ , let  $q \in \text{int } \Delta(\mathcal{Y})$  be the current reference marginal and let  $\mathfrak{B}_t(s)$  be a calibrated convex set of pair beliefs  $\nu$  on  $(\omega, y)$ , supported on  $y = \chi(\omega)$ . For each  $\nu$ , write  $\nu_\chi$  for its index marginal and define

$$\ell_\nu(s, p) = \mathbb{E}_{(\omega, y) \sim \nu} \ell_\omega(s, p), \quad \mathcal{I}_\chi(s, p; \nu) = \mathbb{E}_{a \sim p} \mathcal{I}_\nu(Y; O \mid s, a).$$

With coefficient  $\gamma > 0$ , the one-step robust AIR/EBO objective is

$$\mathfrak{A}_{q, \gamma}(s, p, \nu) := \ell_\nu(s, p) - \gamma \mathcal{I}_\chi(s, p; \nu) - \gamma D_{\text{KL}}(\nu_\chi \| q). \quad (45)$$

The robust maximization over beliefs is essential:

$$p_t \in \arg \min_{p \in \Delta(\mathcal{A}(s))} \max_{\nu \in \mathfrak{B}_t(s)} \mathfrak{A}_{q_t, \gamma}(s, p, \nu), \quad \bar{\nu}_t \in \arg \max_{\nu \in \mathfrak{B}_t(s)} \mathfrak{A}_{q_t, \gamma}(s, p_t, \nu). \quad (46)$$

Without the inner  $\max_\nu$ , the display is only a posterior AIR calculation for a chosen belief, not a robust AMS/EBO rule.

The convex-analytic interpretation is the following. The fixed-truth proof uses the coordinate log loss  $\gamma \log(1/q_t(y^*))$ . Averaging this coordinate under a candidate belief gives  $\gamma H(\nu_\chi) + \gamma D_{\text{KL}}(\nu_\chi \| q_t)$ . When the next reference marginal is the posterior coordinate induced by  $\nu$ , the expected change of this averaged potential is  $-\gamma \mathcal{I}_\chi(s, p; \nu) - \gamma D_{\text{KL}}(\nu_\chi \| q_t)$ . Equivalently, optimizing over possible next index laws yields the log-partition dual (33). Hence the EBO potential is the KL-dual log

partition, while the fixed-truth AIR certificate is the coordinate posterior log loss; they coincide only through KL duality and the chosen update, not as identical scalar functions.

For a genuine convex belief optimization one must verify the convexity/concavity assumptions used by the saddle argument. In finite exponential-weights or admissible-relaxation settings this follows from the log-partition dual. In general model classes, mutual information need not be concave in an arbitrary belief when both the index marginal and the observation kernels vary. One must either verify concavity of  $\nu \mapsto \mathfrak{A}_{q,\gamma}(s, p, \nu)$  on  $\mathfrak{B}_t(s)$ , or replace it by a certified concave upper relaxation.

**Lemma 4.6** (AMS/EBO saddle certifies the AIR fixed-truth bracket). *Assume the finite or dominated differentiability setting of Lemma 3.4. Fix  $(\omega^*, y^*)$  with  $y^* = \chi(\omega^*)$ . Suppose  $\nu \mapsto \mathfrak{A}_{q_t,\gamma}(s, p_t, \nu)$  is concave on the convex set  $\mathfrak{B}_t(s)$ ,  $\bar{\nu}_t$  maximizes it as in (46), and the comparison direction  $\delta_{(\omega^*, y^*)} - \bar{\nu}_t$  is admissible for  $\mathfrak{B}_t(s)$ ; any failure of this containment is charged as a calibration error. If*

$$\max_{\nu \in \mathfrak{B}_t(s)} \mathfrak{A}_{q_t,\gamma}(s, p_t, \nu) \leq \varepsilon_t,$$

*then the fixed-truth AIR bracket with coefficient  $\gamma$  satisfies*

$$\ell_{\omega^*}(s, p_t) - \gamma \mathbb{E}_{a \sim p_t, O \sim P_{\omega^*, s, a}} \log \frac{\bar{\nu}_t(Y = y^* \mid s, a, O)}{q_1(y^*)} \leq \varepsilon_t. \quad (47)$$

*If the reference update is  $q_{t+1}(\cdot) = \bar{\nu}_t(Y \in \cdot \mid s, A_t, O_t)$  and the calibration assumptions hold along the trajectory, then Theorem 4.1 with  $V_t \equiv 0$  gives*

$$\mathbb{E}_{\omega^*} \text{Reg}_T \leq \gamma \log \frac{1}{q_1(y^*)} + \sum_{t=1}^T \varepsilon_t + \text{cal}_T \quad (48)$$

*for finite or countable index sets. Averaging this coordinate bound under a Bayesian prior replaces  $\log(1/q_1(Y))$  by the corresponding cross-entropy or by  $H_\mu(Y)$  when  $q_1 = \mu_\chi$ .*

*Proof.* The AIR gradient calculation in Lemma 3.4, with  $\gamma = 1/\eta$ , identifies the fixed-truth bracket as the first-order value of  $\mathfrak{A}_{q_t,\gamma}$  in the direction  $\delta_{(\omega^*, y^*)} - \bar{\nu}_t$ . Concavity and optimality of  $\bar{\nu}_t$  make that first-order correction nonpositive for every admissible comparison direction. Hence the bracket is no larger than  $\mathfrak{A}_{q_t,\gamma}(s, p_t, \bar{\nu}_t)$ , which is at most the robust value and therefore at most  $\varepsilon_t$ . The regret bound is the fixed-truth log-potential telescope.  $\square$

A dynamic AMS/EBO relaxation is obtained by adding a continuation term before the robust maximization:

$$\mathfrak{A}_{t,q,\gamma}^V(s, p, \nu) = \ell_\nu(s, p) + \mathbb{E}_\nu V_{t+1}(S^+) - V_t(s) - \gamma \mathcal{I}_\chi(s, p; \nu) - \gamma D_{\text{KL}}(\nu_\chi \| q).$$

This is a valid relaxation of the robust Bellman recursion (35) only after the same first-order or direct-bracket certificate is proved. The standard one-step EBO display is the case  $V_t \equiv 0$ ; the dynamic version is an admissible relaxation of the central log-penalized Bellman program, not an automatic consequence of the one-step objective.

**Remark 4.7** (The one identity and the four families). *The identity is the fixed-truth log-potential telescope (29). The exact information-risk Bellman program is (35). Posterior-averaged Bayes analysis is obtained by averaging the same coordinate identity, giving (37). UCB certifies the bracket through confidence and optimism. E2D/DEC replaces the continuation by a one-step separation penalty. AMS/EBO optimizes a KL-dual robust belief relaxation, with the required  $\max_\nu$  appearing explicitly. Thus the common object is not a universal one-step coefficient but the logarithmic Bellman bracket on a sufficient state and a chosen index.*

## 5 Lower bounds: reference histories and quantile indices

### 5.1 Ghost probability and true good probability

Fix an average-regret threshold  $r \geq 0$ . The posterior-reference ghost-good probability is

$$p_r^{\text{Alg}}(\mu, \chi) := \mathbb{P}_{Y \sim \mu_\chi, H'_T \sim \bar{\mathbb{P}}_\mu^{\text{Alg}}} (\bar{L}_\chi(Y, H'_T) \leq r), \quad (49)$$

where  $Y$  and  $H'_T$  are independent under the ghost law and  $\mu_\chi$  is the marginal law of  $Y$ . The true good probability is

$$q_r^{\text{Alg}}(\mu, \chi) := \mathbb{P}_{Y, H_T} (\bar{L}_\chi(Y, H_T) \leq r), \quad (50)$$

where  $(Y, H_T)$  are generated by  $\Omega \sim \mu$  and  $H_T \sim \mathbb{P}_\Omega^{\text{Alg}}$ .

The ghost probability is algorithm-dependent. That dependence is intentional: it preserves the reference trajectory geometry instead of replacing it with a worst-case static small ball.

### 5.2 Reference-history quantile theorem

Let

$$\text{kl}(q, p) = q \log \frac{q}{p} + (1 - q) \log \frac{1 - q}{1 - p}$$

be binary KL divergence. For  $p \in [0, 1]$  and  $c \geq 0$ , define

$$\psi(p, c) := \sup\{q \in [0, 1] : \text{kl}(q, p) \leq c\}.$$

**Theorem 5.1** (Reference-history quantile indexed-information lower bound). *For every prior  $\mu$ , index map  $\chi$ , threshold  $r \geq 0$ , and algorithm  $\text{Alg}$ ,*

$$\text{kl}(q_r^{\text{Alg}}(\mu, \chi), p_r^{\text{Alg}}(\mu, \chi)) \leq C_\chi^{\text{Alg}}(\mu) = T \bar{C}_\chi^{\text{Alg}}(\mu). \quad (51)$$

Consequently,

$$\mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) \geq Tr \left[ 1 - \psi(p_r^{\text{Alg}}(\mu, \chi), T \bar{C}_\chi^{\text{Alg}}(\mu)) \right]. \quad (52)$$

In particular, if

$$T \bar{C}_\chi^{\text{Alg}}(\mu) \leq \text{kl}\left(\frac{1}{2}, p_r^{\text{Alg}}(\mu, \chi)\right), \quad (53)$$

then

$$\mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} \bar{L}_\Omega(H_T) \geq \frac{r}{2}, \quad \mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) \geq \frac{Tr}{2}. \quad (54)$$

A convenient sufficient condition is

$$T \bar{C}_\chi^{\text{Alg}}(\mu) + \log 2 \leq \frac{1}{2} \log \frac{1}{p_r^{\text{Alg}}(\mu, \chi)}. \quad (55)$$

*Proof.* Consider the true joint law of  $(Y, H_T)$  and the ghost law of  $(Y, H'_T)$ . Both have the same marginal law of  $Y$ , while  $Y$  and  $H'_T$  are independent under the ghost law. Hence

$$D_{\text{KL}}(\mathcal{L}(Y, H_T) \| \mathcal{L}(Y, H'_T)) = I_\mu(Y; H_T).$$

Data processing applied to the indicator of the event  $\{\bar{L}_\chi(Y, \cdot) \leq r\}$  gives

$$\text{kl}(q_r^{\text{Alg}}, p_r^{\text{Alg}}) \leq I_\mu(Y; H_T).$$

Proposition 3.2 identifies the right-hand side with  $C_\chi^{\text{Alg}}(\mu) = T\bar{C}_\chi^{\text{Alg}}(\mu)$ , proving (51). The definition of  $\psi$  gives  $q_r^{\text{Alg}} \leq \psi(p_r^{\text{Alg}}, T\bar{C}_\chi^{\text{Alg}})$ . By nonnegativity of loss and by (3),

$$\begin{aligned}\mathbb{E}_{\Omega, H_T} L_\Omega(H_T) &= \mathbb{E}_{Y, H_T} L_\chi(Y, H_T) \\ &\geq Tr \mathbb{P}\{\bar{L}_\chi(Y, H_T) > r\} \\ &= Tr(1 - q_r^{\text{Alg}}),\end{aligned}$$

which proves (52). If (53) holds, then  $q_r^{\text{Alg}} \leq 1/2$ , giving (54). Finally,

$$\text{kl}\left(\frac{1}{2}, p\right) = \frac{1}{2} \log \frac{1}{4p(1-p)} \geq \frac{1}{2} \log \frac{1}{p} - \log 2,$$

so (55) implies (53). □

**Relation to interactive Fano.** Theorem 5.1 should be viewed as an indexed, posterior-reference specialization of the interactive Fano method of Chen et al. (2024). The binary-KL step is the same ghost-data argument: one compares the true experiment with a reference experiment, applies data processing to the small-regret event, and obtains a Bayes risk lower bound when the trajectory information is smaller than the logarithmic inverse ghost-good probability. The additional point here is representational. We choose the reference experiment to be the Bayesian posterior-predictive trajectory and choose an information index  $Y = \chi(\Omega)$ . With this choice, the trajectory information admits the exact stepwise identity

$$I_\mu(Y; H_T) = \mathbb{E}_{H'_T \sim \bar{\mathbb{P}}_\mu^{\text{Alg}}} \sum_{t=1}^T \mathcal{I}_\chi(S'_t, p'_t).$$

Thus the same reference history  $H'_T$  generates both the ghost quantile and the indexed information telescope. This is the lower-bound counterpart of the AIR/MAIR upper-bound identity. In particular,  $\chi(\Omega) = \Omega$  gives the model-index MAIR form, while  $\chi(\Omega) = \pi^*(\Omega)$  gives the action-index AIR form.

The novelty is not a new Fano inequality, but a new representation of interactive Fano in posterior-reference indexed-information coordinates. This avoids prematurely collapsing the adaptive experiment to a one-round decision distribution, the step that creates the localized change-of-measure difficulty in one-round DEC reductions. It also makes the lower-bound critical radius directly comparable to AIR/MAIR supersolution upper bounds, since both sides are expressed as sums of the same indexed information along posterior-reference histories.

**Remark 5.2** (Why average notation is useful). *The theorem could be written entirely with cumulative quantities. Writing  $\bar{C}_\chi = T^{-1}\mathbb{E} \sum_t \mathcal{I}_\chi$  and  $\bar{L} = T^{-1}L$  makes the scaling visible:*

$$\text{information side} = T\bar{C}_\chi, \quad \text{regret side} = Tr.$$

*Thus  $T$  appears exactly as in one-round reductions, but no posterior trajectory has been collapsed into a static action distribution.*

### 5.3 Bellman-Fano lower certificates

Theorem 5.1 is algorithm-specific. We now give algorithm-uniform certificates. The first bounds the information of any algorithm; the second bounds the ghost probability of success of any algorithm.



**Information supersolution.** For a function  $F$  on states, define

$$(\mathbf{C}_t F)(s) := \sup_{p \in \Delta(\mathcal{A}(s))} \left\{ \mathcal{I}_\chi(s, p) + \mathbb{E}_{a \sim p, o \sim P_{s,a}} F(\tau(s, a, o)) \right\}.$$

A sequence  $\overline{C}_t$  is an information supersolution if

$$\overline{C}_{T+1} \geq 0, \quad \overline{C}_t(s) \geq (\mathbf{C}_t \overline{C}_{t+1})(s) \quad \forall t, s.$$

Then, for every algorithm starting from  $s_1$ ,

$$C_\chi^{\text{Alg}}(\mu) \leq \overline{C}_1(s_1). \quad (56)$$

The proof is the same backward induction as Proposition 2.2.

**Ghost-good-mass supersolution.** Let  $m \in \Delta(\mathcal{Y})$  be the target-index distribution used to evaluate success, let  $s$  be the posterior-reference state, and let  $g : \mathcal{Y} \rightarrow [0, L]$  be an accumulated average-loss profile. For threshold  $r$ , define the exact recursion

$$\begin{aligned} \Gamma_{T+1}^r(m, s, g) &= m\{y : g(y) \leq r\}, \\ \Gamma_t^r(m, s, g) &= \sup_{p \in \Delta(\mathcal{A}(s))} \mathbb{E}_{a \sim p, o \sim P_{s,a}} \Gamma_{t+1}^r \left( m, \tau(s, a, o), g + \frac{1}{T} \ell^\chi(s, a) \right), \end{aligned}$$

where  $\ell^\chi(s, a)$  is the function  $y \mapsto \ell_y^\chi(s, a)$ . If  $\overline{\Gamma}_t^r$  is a supersolution of this recursion, then for every algorithm,

$$p_r^{\text{Alg}}(\mu, \chi) \leq \overline{\Gamma}_1^r(\mu_\chi, s_1, 0). \quad (57)$$

Define the ghost entropy lower certificate

$$\underline{\mathcal{E}}_1^r(\mu, \chi) := -\log \overline{\Gamma}_1^r(\mu_\chi, s_1, 0). \quad (58)$$

**Theorem 5.3** (Bellman quantile-index lower certificate). *Suppose  $\overline{C}_t$  is an information supersolution and  $\overline{\Gamma}_t^r$  is a ghost-good-mass supersolution. Then*

$$\inf_{\text{Alg}} \mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) \geq Tr \left[ 1 - \psi(e^{-\underline{\mathcal{E}}_1^r(\mu, \chi)}, \overline{C}_1(s_1)) \right]. \quad (59)$$

In particular, if

$$\overline{C}_1(s_1) + \log 2 \leq \frac{1}{2} \underline{\mathcal{E}}_1^r(\mu, \chi), \quad (60)$$

then

$$\inf_{\text{Alg}} \mathbb{E}_{\Omega \sim \mu} \mathbb{E}_{\mathbb{P}_\Omega^{\text{Alg}}} L_\Omega(H_T) \geq \frac{Tr}{2}. \quad (61)$$

Consequently,

$$\mathfrak{R}_T^* \geq \sup_{\mu, \chi, r} \left\{ \frac{Tr}{2} : \overline{C}_1(s_1) + \log 2 \leq \frac{1}{2} \underline{\mathcal{E}}_1^r(\mu, \chi) \right\}.$$

*Proof.* For every algorithm, (56) gives  $T \overline{C}_\chi^{\text{Alg}}(\mu) = C_\chi^{\text{Alg}}(\mu) \leq \overline{C}_1(s_1)$ , and (57) gives  $p_r^{\text{Alg}}(\mu, \chi) \leq e^{-\underline{\mathcal{E}}_1^r(\mu, \chi)}$ . Substitute these two inequalities into Theorem 5.1, using the monotonicity of  $\psi$  in both arguments. The minimax statement follows from Yao's principle.  $\square$

The critical average-regret radius is

$$r_\star(\mu, \chi) := \sup \left\{ r : \overline{C}_1(s_1) + \log 2 \leq \frac{1}{2} \underline{\mathcal{E}}_1^r(\mu, \chi) \right\}. \quad (62)$$

The lower bound is order  $Tr_\star$ . A two-point Le Cam proof corresponds to the special case in which  $\underline{\mathcal{E}} = O(1)$ . A Fano or local-prior-mass lower bound has  $\underline{\mathcal{E}}$  of order dimension or effective dimension.

## 5.4 DEC as one-step relaxation

As discussed in Section 4, offset DEC is obtained from the fixed-truth log-potential bracket (28) by making a one-step relaxation: the coordinate potential drop, or its posterior-averaged Bellman continuation counterpart, is replaced by a static one-step separation penalty, and the posterior-reference law is replaced by a fixed one-step reference predictive law.

In the model-index case  $\chi(\omega) = \omega$ , let  $\bar{\nu}$  be a reference belief at state  $s$ , and write

$$P_{\bar{\nu},s,a} := \int P_{\omega,s,a} \bar{\nu}(d\omega)$$

for its predictive law. The corresponding pointwise KL offset is

$$\text{dec}_{\gamma}^{\text{KL}}(s; \bar{\nu}) := \inf_{p \in \Delta(\mathcal{A}_s)} \sup_{\omega \in \Omega_s} \{ \mathbb{E}_{a \sim p} \ell_{\omega}(s, a) - \gamma \mathbb{E}_{a \sim p} D_{\text{KL}}(P_{\omega,s,a} \| P_{\bar{\nu},s,a}) \}. \quad (\text{one-step model-index offset})$$

Thus the usual offset DEC is a worst-case, static-reference relaxation of the Bellman program: the learner chooses the one-step decision distribution  $p$ , and the local adversary then chooses the hardest model relative to the fixed reference predictive law.

The same one-step logic also appears on the lower-bound side, but only after an additional reduction from a dynamic posterior-reference experiment to a one-step reference experiment. Theorem 5.1 and Theorem 5.3 retain the dynamic reference posteriors. A one-step quantile DEC is obtained only by relaxing this dynamic proof: one fixes a reference model, or more generally a reference predictive law, decomposes the trajectory divergence into one-step divergences, and replaces the algorithm-dependent ghost history by a one-round decision distribution. In Chen et al. (2024, Section 3.2.2), this idea is made explicit through the quantile PAC-DEC. A reference model  $\bar{M}$  generates the ghost experiment, an adversarial model  $M$  supplies the hard alternative, and the loss certificate is a quantile of the one-step risk rather than its expectation. The further reduction to constrained regret-DEC in that paper is precisely where localization, regularity, and problem-specific assumptions enter.

The passage from the dynamic theorem to a one-step coefficient is therefore not merely notational. On both the upper- and lower-bound sides, existing DEC measures can be viewed as one-step relaxations obtained by freezing the posterior-reference process into reference models or predictive laws. This interpretation is primarily conceptual: even the distinction between quantile and expected loss raises nontrivial technical questions when deriving regret-DEC lower bounds under regularity conditions. Nevertheless, the present paper adopts this viewpoint because it clarifies how one-step DEC certificates arise from the dynamic indexed Bellman picture.

Thus DEC/E2D should be read as a one-step relaxation of indexed Bellman information complexity, rather than as a universally tight conversion coefficient from information to regret. When the posterior geometry is simple, this relaxation can be sharp. In more structured problems, however, a one-round coefficient can obscure the dynamic object in two distinct ways: it may charge an estimation or entropy cost that the posterior-reference lower bound has already paid, or it may discard the geometry of the reference trajectory that enables localization. The examples in Section 6 illustrate indexed-Bellman certificates and diagnostics; the following informal diagnostics explain why the corresponding one-round DEC accounting should be interpreted with care.

- *Vacuity for RKHS classes without localization.* In infinite-dimensional kernel bandits, a global one-round DEC over an RKHS ball can be unbounded or vacuous unless the comparison is localized to the finite active experiment actually visited by the algorithm; see the end of Section 6.1 for details. Known routes to matching DEC-style lower bounds introduce this

localization explicitly or impose regularity conditions that enforce it implicitly through a critical-radius or sub-root fixed point (Foster et al., 2023). Indexed Bellman information complexity keeps this localization inside the posterior-reference process itself.

- *Double counting in multi-armed bandits.* The hard prior and the ghost entropy already contain the cost of identifying one arm among  $K$ . Therefore, after a Fano or posterior-reference calculation has produced the logarithmic entropy term, appending an independent DEC “estimation” penalty charges the same information budget a second time. This is not a statement that DEC-based algorithms are necessarily suboptimal for multi-armed bandits; it is a statement that the entropy appearing in the lower certificate and the estimation term appearing in a one-step upper relaxation should not be added as independent costs.
- *Collapsed lower bounds in linear bandits.* A one-round coefficient obtained by collapsing the adaptive trajectory to a single action distribution can leave a localization or logarithmic gap in the lower bound. The gap is not intrinsic to the linear-bandit experiment: it is introduced by replacing the posterior trajectory by a static action law. The indexed Bellman formulation keeps the posterior covariance, the information telescope, and the ghost quantile on the same reference path.
- *Representational ambiguity for Bellman rank.* Bellman rank is a representation-relative algebraic certificate: it depends on a hypothesis class, value-function class, roll-in distribution, discrepancy, and Bellman-error witness (Jiang et al., 2017). Without fixing these observable witnesses, bounded Bellman rank is not itself a DMSO model class and does not determine an intrinsic DEC. Once a bilinear-class witness and its estimation complexity are specified, DEC-style questions become meaningful (Du et al., 2021). Thus, a lower bound requires an explicit hard family, observable witnesses, and a ghost-entropy calculation within the specified class.

**Definition 5.4** (Tight indexed Bellman representation). *Fix a problem family, a horizon sequence, and a class of admissible indices  $\chi$ . An indexed Bellman representation is tight if, for the relevant initial states  $s_1$  and for some prior-index pair  $(\mu, \chi)$ , there exist a regret supersolution  $\bar{V}$  and Bellman-Fano lower certificates  $\bar{C}, \underline{\mathcal{E}}$  on the same state space such that, up to the stated universal constants or logarithmic factors,*

$$\bar{V}_1(s_1) \lesssim Tr_\star(\mu, \chi), \quad \mathfrak{R}_T^\star \gtrsim Tr_\star(\mu, \chi),$$

where  $r_\star(\mu, \chi)$  is the critical average-regret radius in (62).

This definition is the representation-level replacement for asking whether a one-round DEC coefficient is universally tight. The question is not whether a single static coefficient is always exact, but whether the same Bellman-sufficient state supports both an upper supersolution and a posterior-reference quantile lower certificate at the same critical radius.

## 6 Applications and examples

Before turning to examples, we spell out what the state means in common problems. In DMSO, the state may be the posterior over models, with the full history as a fallback sufficient statistic. In multi-armed bandits, the state can be the vector of posterior sufficient statistics for arm means, and the natural AIR index is the optimal arm. In linear bandits, a Gaussian or confidence-state representation consists of a posterior mean and covariance, or their frequentist confidence analogues;

the index may be the parameter, an optimal action, or a coarse optimal direction. In finite-action kernel bandits, the state is the active marginal posterior on the finite action set; the unknown function remains an RKHS element. In low-Bellman-rank problems, the state must include the witness features and value-function information needed to predict Bellman errors. Thus “state” never means a universal finite-dimensional object. It means a representation, preferably minimal but at least sufficient, on which the Bellman, information, and ghost-good recursions close.

### 6.1 Finite-action kernel bandits: indexed Bellman DP, GP-UCB, GP-E2D, and AMS/EBO

This subsection specializes the indexed Bellman language to kernel bandits on a finite active action set. The point is not to make the unknown function finite-dimensional. The truth remains an element of an RKHS. What becomes finite is the marginal experiment observed through the active actions. This finite marginal is enough to define the GP posterior update, the model-index MAIR information, and the action-index AIR belief over the optimal active action. We present the exact finite-marginal indexed Bellman program first, and then compare GP-UCB, GP-E2D, and AMS/EBO as tractable relaxations or certificates of its bracket.

**Problem and finite active marginal.** Let  $\mathcal{H}_k$  be a reproducing kernel Hilbert space (RKHS) on a possibly infinite domain  $\mathcal{X}_{\text{all}}$ , with kernel  $k$  satisfying  $k(x, x) \leq \kappa^2$ . The learner is evaluated on a finite active set

$$\mathcal{X} = \{x_1, \dots, x_n\} \subset \mathcal{X}_{\text{all}}.$$

The unknown reward function  $f^* \in \mathcal{H}_k$  satisfies  $\|f^*\|_{\mathcal{H}_k} \leq B$ , and at round  $t$  the learner chooses  $X_t \in \mathcal{X}$  and observes

$$Y_t = f^*(X_t) + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid H_{t-1}, X_t] = 0, \quad (63)$$

where the noise is conditionally  $R$ -sub-Gaussian. For the Gaussian calculations below we use the algorithmic reference model with variance parameter  $\lambda > 0$ ; the frequentist confidence event is obtained by the usual self-normalized argument.

Let  $F = (f(x_1), \dots, f(x_n)) \in \mathbb{R}^n$  be the active reward vector and let  $K_{\mathcal{X}}$  be the  $n \times n$  kernel matrix. The GP reference prior is

$$F \sim N(0, K_{\mathcal{X}}), \quad Y_t \mid F, X_t = x_i \sim N(F_i, \lambda).$$

After history  $H_{t-1}$ , the algorithmic posterior on  $F$  is Gaussian,

$$F \mid H_{t-1} \sim N(m_t, \Sigma_t).$$

For  $x_i \in \mathcal{X}$ , write  $m_t(x_i) = e_i^\top m_t$  and  $s_t^2(x_i) = e_i^\top \Sigma_t e_i$ . If  $X_t = x_i$ , the exact posterior update is

$$m_{t+1} = m_t + \frac{\Sigma_t e_i}{\lambda + s_t^2(x_i)} \{Y_t - m_t(x_i)\}, \quad (64)$$

$$\Sigma_{t+1} = \Sigma_t - \frac{\Sigma_t e_i e_i^\top \Sigma_t}{\lambda + s_t^2(x_i)}. \quad (65)$$

This is a finite marginal posterior update. It should not be read as an assumption that  $f^*$  has  $n$  parameters. The RKHS function may be infinite-dimensional; only the observations and decisions in this finite experiment depend on the vector  $F$ .

**Model-index information and action-index information.** For the model-index specialization, take the index to be the active reward vector  $Y_{\text{MAIR}} = F$ . The one-step GP/MAIR information at state  $H_{t-1}$  is

$$\mathcal{I}_t^{\text{GP}}(p) := \mathbb{E}_{x \sim p} \frac{1}{2} \log \left( 1 + \frac{s_t^2(x)}{\lambda} \right). \quad (66)$$

Along a realized action sequence,

$$C_T^{\text{GP}} := \sum_{t=1}^T \mathcal{I}_t^{\text{GP}}(\delta_{X_t}) = \frac{1}{2} \log \det(I + \lambda^{-1} K_{X_{1:T}, X_{1:T}}), \quad (67)$$

where repeated actions are included in the Gram matrix. This is the realized information gain of the algorithmic trajectory. It can be much smaller than the maximal information gain that maximizes over all length- $T$  designs.

For the action-index specialization, assume ties are broken by a fixed rule and set

$$A^*(F) \in \arg \max_{x \in \mathcal{X}} F_x, \quad q_t(a) := \mathbb{P}(A^* = a \mid H_{t-1}).$$

Let  $\nu_t^a$  be the exact conditional law of  $F$  given  $H_{t-1}$  and  $A^* = a$ . It is generally a truncated Gaussian on the cone where  $a$  is optimal. Define the conditional predictive law

$$P_{t,a,x}(\cdot) := \int N(F_x, \lambda)(\cdot) \nu_t^a(dF),$$

for the observation at action  $x$  given the event  $A^* = a$ . The marginal predictive is

$$P_{t,x} = \sum_{a \in \mathcal{X}} q_t(a) P_{t,a,x} = N(m_t(x), \lambda + s_t^2(x)).$$

The AIR information increment is

$$\mathcal{I}_t^{\text{AIR}}(p) := \mathbb{E}_{x \sim p} \sum_{a \in \mathcal{X}} q_t(a) D_{\text{KL}}(P_{t,a,x} \| P_{t,x}). \quad (68)$$

The conditional AIR regret is

$$\Delta_t^{\text{AIR}}(p) := \sum_{a \in \mathcal{X}} q_t(a) \mathbb{E}_{F \sim \nu_t^a} \mathbb{E}_{x \sim p} [F_a - F_x]. \quad (69)$$

The exact indexed chain rule gives

$$\sum_{t=1}^T \mathbb{E} \mathcal{I}_t^{\text{AIR}}(p_t) = I(A^*; H_T) \leq H(A^*) \leq \log n.$$

Thus AIR places the entropy telescope on the decision index rather than on the ambient RKHS. The price is that the coefficient converting action-index information into regret must be controlled by an algorithmic bracket.

**Algorithm 1: finite-marginal AIR/MAIR Bellman DP.** On the finite active marginal, the exact frequentist benchmark is the log-penalized Bellman program of Theorem 4.2. The state contains  $(m_t, \Sigma_t, q_t)$ , where  $(m_t, \Sigma_t)$  is the algorithmic GP marginal on  $F$  and  $q_t$  is the maintained reference law of the index, either  $A^*(F)$  for AIR or  $F$  for MAIR. Let  $\mathfrak{C}_t(m, \Sigma, q)$  be a calibrated local set of candidate active reward vectors containing the fixed truth  $F^*$  on the calibration event. In the action-index version define

$$y(f) = A^*(f), \quad \mathbf{G}_t^{\text{AIR}}(p; f) := \mathbb{E}_{x \sim p, O \sim N(f_x, \lambda)} \log \frac{q^+(y(f) \mid m, \Sigma, q, x, O)}{q(y(f))},$$

where  $q^+$  is the chosen reference update for the optimal-action marginal. The robust finite-marginal Bellman recursion is

$$W_t^\gamma(m, \Sigma, q) = \inf_{p \in \Delta(\mathcal{X})} \sup_{f \in \mathfrak{C}_t(m, \Sigma, q)} \left\{ \Delta_f(p) + \mathbb{E}_f W_{t+1}^\gamma(m^+, \Sigma^+, q^+) - \gamma \mathbf{G}_t^{\text{AIR}}(p; f) \right\}.$$

The model-index MAIR version replaces  $y(f)$  by the finite reward vector  $f$  and uses the corresponding model-coordinate posterior log gain. By Theorem 4.2, an approximate minimizer of this recursion has regret at most

$$W_1^\gamma(m_1, \Sigma_1, q_1) + \gamma \log \frac{1}{q_1(A^*(F^*))} + \text{optimization and calibration errors}.$$

If the maintained belief is the exact Bayesian posterior and one averages this coordinate recursion over  $F$ , the robust supremum is replaced by posterior expectation and the dynamic program reduces to the posterior-averaged AIR recursion

$$U_t^\gamma(m, \Sigma, q) = \inf_{p \in \Delta(\mathcal{X})} \left\{ \Delta_t^{\text{AIR}}(p) + \mathbb{E} U_{t+1}^\gamma(m^+, \Sigma^+, q^+) - \gamma \mathcal{I}_t^{\text{AIR}}(p) \right\}.$$

Thus the finite-marginal Bellman DP is the clean upper object for comparison with the lower theorem: the frequentist version uses the coordinate log potential and calibrated candidate truths, while the Bayesian version is its posterior average. The following three rules are tractable certificates or relaxations of this benchmark.

**Algorithm 2: GP-UCB.** GP-UCB uses the Gaussian marginal posterior  $(m_t, \Sigma_t)$  on the finite active marginal. In this paragraph,  $e_x$  denotes the coordinate vector associated with  $x \in \mathcal{X}$ , and  $s_t(x) = (e_x^\top \Sigma_t e_x)^{1/2}$  denotes the unmultiplied posterior standard deviation, i.e., the generic uncertainty scale  $\sigma_t(x)$  from Lemma 4.5. The confidence half-width is

$$w_t(x) := \beta_t s_t(x),$$

where  $\beta_t$  is the confidence multiplier. The algorithm chooses

$$X_t \in \arg \max_{x \in \mathcal{X}} \{m_t(x) + \beta_t s_t(x)\}. \quad (70)$$

The following event is the frequentist calibration certificate:

$$\mathcal{E}_{\text{GP}} := \{|f^*(x) - m_t(x)| \leq w_t(x) = \beta_t s_t(x) \text{ for all } t \leq T, x \in \mathcal{X}\}. \quad (71)$$

For finite  $\mathcal{X}$ , standard RKHS self-normalized concentration gives  $\mathbb{P}(\mathcal{E}_{\text{GP}}) \geq 1 - \delta$  for the usual choice of  $\beta_t$ . For example, up to universal constants one may take

$$\beta_t \asymp B + R \sqrt{\Gamma_{t-1}^{\text{GP}} + \log(1/\delta)}, \quad \Gamma_m^{\text{GP}} := \sup_{x_{1:m} \in \mathcal{X}^m} \frac{1}{2} \log \det(I + \lambda^{-1} K_{x_{1:m}, x_{1:m}}),$$

with the precise form depending on the normalization of  $k$  and the ridge parameter. Here  $K_{x_{1:m}, x_{1:m}}$  is the Gram matrix of the possibly repeated design sequence  $(x_1, \dots, x_m)$ . The quantity  $\Gamma_m^{\text{GP}}$  is a worst-case calibration device for the confidence event; it is different from the realized information gain  $C_T^{\text{GP}}$  paid in the regret bound. The next theorem isolates the deterministic part of the GP–UCB proof.

**Theorem 6.1** (GP–UCB realized-information bound). *On the event  $\mathcal{E}_{\text{GP}}$ , the GP–UCB rule (70) satisfies*

$$\text{Reg}_T(f^*) \leq 2\beta_T \sum_{t=1}^T s_t(X_t) \leq 2\beta_T \sqrt{T c_{\kappa, \lambda} C_T^{\text{GP}}}, \quad (72)$$

where

$$c_{\kappa, \lambda} := \frac{2\kappa^2}{\log(1 + \kappa^2/\lambda)}$$

and  $C_T^{\text{GP}}$  is the realized information gain in (67). Consequently, if rewards are bounded in  $[0, 1]$  and  $\mathbb{P}(\mathcal{E}_{\text{GP}}) \geq 1 - \delta$ , then

$$\mathbb{E} \text{Reg}_T(f^*) \leq 2\beta_T \sqrt{T c_{\kappa, \lambda} \mathbb{E} C_T^{\text{GP}}} + T\delta.$$

*Proof.* Let  $x^* \in \arg \max_{x \in \mathcal{X}} f^*(x)$ . On  $\mathcal{E}_{\text{GP}}$ , calibration at  $x^*$  and the optimistic choice of  $X_t$  give

$$f^*(x^*) \leq m_t(x^*) + w_t(x^*) \leq m_t(X_t) + w_t(X_t).$$

Calibration at  $X_t$  gives  $m_t(X_t) \leq f^*(X_t) + w_t(X_t)$ . Hence

$$f^*(x^*) - f^*(X_t) \leq 2w_t(X_t) = 2\beta_t s_t(X_t) \leq 2\beta_T s_t(X_t).$$

Summing and applying Cauchy–Schwarz yields

$$\text{Reg}_T(f^*) \leq 2\beta_T \sqrt{T \sum_{t=1}^T s_t^2(X_t)}.$$

For  $u \in [0, \kappa^2/\lambda]$ , monotonicity of  $u/\log(1+u)$  on a bounded interval gives

$$\lambda u \leq \frac{\kappa^2}{\log(1 + \kappa^2/\lambda)} \log(1 + u).$$

Applying this with  $u = s_t^2(X_t)/\lambda$  and using (67) proves (72). The expectation bound adds the trivial regret bound  $T$  on  $\mathcal{E}_{\text{GP}}^c$  and uses Jensen’s inequality.  $\square$

**GP–UCB as a MAIR decomposition.** The theorem is Lemma 4.5 with the generic uncertainty scale  $\sigma_t(X_t) = s_t(X_t)$  and the realized-information log determinant telescope

$$\sum_t \frac{1}{2} \log(1 + s_t^2(X_t)/\lambda) = C_T^{\text{GP}}.$$

The confidence half-width is  $w_t(X_t) = \beta_t s_t(X_t)$ , but the multiplier  $\beta_t$  is not part of the information telescope; it is the coefficient paid for calibration and optimism in the fixed-truth bracket. This separation is the reason the proof uses both  $s_t$  and  $w_t$ :  $s_t^2$  is summed by the log-determinant/elliptical-potential certificate, whereas  $w_t$  is used only to upper bound instantaneous regret. GP–UCB is often practically strong because it follows the realized trajectory and pays the realized information gain  $C_T^{\text{GP}}$ , not the worst possible design information  $\Gamma_T^{\text{GP}}$ , even though its confidence multiplier may be conservative in step-uniform minimax analyses.

**Algorithm 3: GP–E2D.** A GP–E2D rule keeps the same GP posterior reference but chooses a distribution over actions by solving a localized one-round MAIR/DEC offset. Given a localized comparison set  $\mathcal{F}_t$  containing plausible functions, define

$$\Delta_f(p) := \max_{x \in \mathcal{X}} f(x) - \mathbb{E}_{x \sim p} f(x),$$

and let  $P_{f,x} = N(f(x), \lambda)$  while  $P_{t,x} = N(m_t(x), \lambda + s_t^2(x))$  is the posterior predictive. A schematic GP–E2D decision is

$$p_t \in \arg \min_{p \in \Delta(\mathcal{X})} \sup_{f \in \mathcal{F}_t} \{ \Delta_f(p) - \gamma_t \mathbb{E}_{x \sim p} D_{\text{KL}}(P_{f,x} \| P_{t,x}) \}. \quad (73)$$

If  $f^* \in \mathcal{F}_t$  and the optimized value in (73) is at most  $\varepsilon_t$ , then

$$\Delta_{f^*}(p_t) \leq \gamma_t \mathbb{E}_{x \sim p_t} D_{\text{KL}}(P_{f^*,x} \| P_{t,x}) + \varepsilon_t. \quad (74)$$

Summing (74) gives a regret bound once the fixed-truth KL terms are related to a posterior-ratio or information telescope. Under a well-specified Bayesian GP reference this relation holds in expectation by the MAIR chain rule. Under a fixed frequentist truth it requires calibration/localization. This is the technical point that a GP–E2D analysis must address: the one-round DEC is optimized, but the compatibility between that coefficient and the realized GP log determinant is not automatic. An unlocalized offset over an infinite-dimensional RKHS ball can be unbounded or vacuous; meaningful GP–E2D guarantees must localize the comparison set to the posterior-reference experiment.

**Algorithm 4: AMS/EBO on the action index.** AMS/EBO attacks the same specialized log-potential bracket from the robust posterior side. In the finite-action kernel marginal, the natural AIR index is  $A^*$ . Let  $q_t$  be the current reference distribution on  $A^*$ , let  $\mathfrak{B}_t(q_t)$  be a calibrated local set of candidate beliefs  $\nu$  on the active reward vector  $F$ , and fix a constant coefficient  $\gamma > 0$  (equivalently  $\eta = 1/\gamma$ ). For each  $\nu$ , define  $\Delta_\nu^{\text{AIR}}(p)$  and  $\mathcal{I}_\nu^{\text{AIR}}(p)$  by the analogues of (69) and (68), using the conditional laws induced by  $\nu$ . A robust AIR/EBO decision is

$$p_t \in \arg \min_{p \in \Delta(\mathcal{X})} \sup_{\nu \in \mathfrak{B}_t(q_t)} \{ \Delta_\nu^{\text{AIR}}(p) - \gamma \mathcal{I}_\nu^{\text{AIR}}(p) - \gamma D_{\text{KL}}(\nu_{A^*} \| q_t) \}. \quad (75)$$

The maximization over  $\nu$  is essential: without it the display is only the posterior AIR bracket for a chosen belief, not the robust AMS/EBO relaxation. In a finite-index abstraction, the last two terms are the KL-dual form of the log-partition potential (33). If the robust value in (75) is at most  $\varepsilon_t$ , the required concavity/first-order conditions in Lemma 4.6 hold, and the belief set is calibrated so that the fixed truth  $f^*$  is an admissible comparison direction up to error  $\text{cal}_T$ , then Theorem 4.1 gives

$$\mathbb{E}_{f^*} \text{Reg}_T \leq \gamma \log \frac{1}{q_1(a^*)} + \mathbb{E}_{f^*} \sum_{t=1}^T \varepsilon_t + \text{cal}_T, \quad (76)$$

where  $a^* = A^*(f^*)$ . The constant coefficient avoids the false shortcut of replacing a weighted fixed-truth log-gain sum by a single telescope. If a varying coefficient is used, the corresponding weighted telescope must be proved separately. Under a Bayesian reference prior, posterior averaging of the same coordinate bound gives

$$\mathbb{E} \text{Reg}_T \leq \gamma H(A^*) + \mathbb{E} \sum_{t=1}^T \varepsilon_t + \mathbb{E} \text{cal}_T.$$



A ratio form gives the familiar Cauchy–Schwarz variant after replacing the fixed coefficient by the appropriate sequential AIR ratio and applying the action-index chain rule. Since  $H(A^*) \leq \log n$ , this route scales with the finite decision index and the sequential AIR coefficient. It is different from GP–E2D: AMS/EBO uses a robust action-index belief trajectory, whereas GP–E2D optimizes a one-round model-index offset whose relationship to the realized log determinant or posterior-ratio telescope must be justified separately.

**Position relative to the GP–UCB optimality debate.** The original GP–UCB analysis is practical and trajectory-sensitive because it pays the realized information gain (67), or a worst-case relaxation only at the final step (Srinivas et al., 2010). Later work asks whether the remaining suboptimality comes from the algorithm or from the confidence analysis; in particular, Vakili et al. (2021b) ask for sharper online RKHS confidence intervals and sharper finite-action kernel-bandit guarantees. The indexed view separates the issue into four mechanisms. The finite-marginal indexed Bellman DP is the exact information-potential benchmark, but may be computationally demanding. GP–UCB has the strongest realized-information geometry but may have a conservative MAIR coefficient. AMS/EBO can keep the action-index AIR telescope  $I(A^*; H_T) \leq \log n$  and therefore has a principled route to finite-decision complexity, provided the robust bracket can be controlled. GP–E2D directly optimizes a one-round DEC coefficient, but an unlocalized DEC can be vacuous and a localized DEC must be matched to the same posterior-reference trajectory. Thus these algorithms should not be compared by a single universal coefficient; they are one exact Bellman benchmark and three relaxations or certificates of the same indexed Bellman identity, with different estimation and coefficient mechanisms.

**Why global RKHS DEC can be vacuous.** Consider a Gaussian RKHS bandit with noise variance  $\sigma^2$ , feature map  $\phi(x) \in \mathcal{H}$  satisfying  $\|\phi(x)\|_{\mathcal{H}} \leq 1$ , and suppose the action set contains points  $\{x_j : j \geq 1\}$  with  $\phi(x_j) = e_j$ , where  $\{e_j\}_{j \geq 1}$  is orthonormal in  $\mathcal{H}$ . Let the reference model be  $f_0 \equiv 0$ , and let  $\mathcal{F}_B = \{f \in \mathcal{H} : \|f\|_{\mathcal{H}} \leq B\}$ . For any one-round decision distribution  $p$ , define

$$q_j := \mathbb{E}_{x \sim p} \langle e_j, \phi(x) \rangle_{\mathcal{H}}^2.$$

Since

$$\sum_{j \geq 1} q_j \leq \mathbb{E}_{x \sim p} \|\phi(x)\|_{\mathcal{H}}^2 \leq 1,$$

there are directions with arbitrarily small  $q_j$ . For the alternative  $f_j = B e_j$ , the optimal action is  $x_j$ , while

$$\mathbb{E}_{x \sim p} \Delta_{f_j}(x) = B - B \mathbb{E}_{x \sim p} \langle e_j, \phi(x) \rangle_{\mathcal{H}} \geq B(1 - \sqrt{q_j}),$$

and the one-round KL information against  $f_0$  is

$$\mathbb{E}_{x \sim p} D_{\text{KL}}(N(f_j(x), \sigma^2) \parallel N(0, \sigma^2)) = \frac{B^2}{2\sigma^2} q_j.$$

Hence, for every finite offset coefficient  $\gamma$ , one may choose  $j$  with  $q_j$  so small that

$$\mathbb{E}_{x \sim p} \Delta_{f_j}(x) - \gamma \mathbb{E}_{x \sim p} D_{\text{KL}}(N(f_j(x), \sigma^2) \parallel N(0, \sigma^2)) \geq cB$$

for a numerical constant  $c > 0$ . The constrained version has the same pathology: for any fixed information radius, a sufficiently rare direction has information inside the radius while retaining regret of order  $B$ . Thus an unlocalized one-round DEC over the entire RKHS ball need not reflect

the effective difficulty of the finite experiment being run. The meaningful object is the finite active marginal together with its posterior-reference trajectory. Indexed Bellman information complexity keeps this localization inside the state process, rather than appending a separate DEC localization after the fact.

**GP–E2D offset must be localized.** The model-index offset in (73) is meaningful only relative to a comparison set and a reference law. If the comparison set is the entire infinite-dimensional RKHS ball, then a one-round adversary can place mass in directions receiving arbitrarily small action probability. The resulting regret remains order one while the one-step information is arbitrarily small, making the global offset vacuous. If the comparison set is instead the finite active posterior neighborhood, the offset may be finite and useful. The point is that the localization is part of the representation and posterior-reference trajectory; it should not be appended as an independent estimation term after a separate log-determinant analysis.

## 6.2 MAB lower bound via optimal-action index and reference history

In this subsection, we apply our lower-bound approach to multi-armed bandits (MAB). Consider  $K \geq 3$  Gaussian arms with unit noise variance. Let the ambient model space be a class  $\Omega_{\text{MAB}} \subseteq \mathbb{R}^K$  of mean vectors, and write a model environment as  $\omega = (\theta_\omega(1), \dots, \theta_\omega(K))$ . Pulling arm  $a$  in environment  $\omega$  produces an observation distributed as  $N(\theta_\omega(a), 1)$ . Let

$$A^*(\omega) \in \arg \max_{a \in [K]} \theta_\omega(a)$$

with an arbitrary fixed tie-breaking rule, and define the information index

$$\chi(\omega) = A^*(\omega) \in [K].$$

Thus the random environment is the full mean vector  $\Omega$ , while the information target used in the lower certificate is only

$$Y = \chi(\Omega) = A^*(\Omega).$$

For the lower bound, fix  $\Delta > 0$  and use the  $K$ -point prior  $\mu_\Delta$  supported on the spike environments

$$\omega^j = \Delta e_j, \quad j \in [K],$$

where  $e_j$  is the  $j$ th coordinate vector. Equivalently, draw  $J \sim \text{Unif}([K])$  and set  $\Omega = \omega^J$ . On this prior support the optimal arm is unique and

$$Y = \chi(\Omega) = J.$$

The equality  $Y = J$  is only a property of this least-favorable prior; it should not be read as identifying the information target with the full model environment. The one-step regret in environment  $\omega^j$  is

$$\ell_{\omega^j}(a) = \max_b \theta_{\omega^j}(b) - \theta_{\omega^j}(a) = \Delta \mathbb{1}\{a \neq j\}.$$

For brevity write  $\ell_j = \ell_{\omega^j}$  on this subfamily.

**Lemma 6.2** (MAB ghost entropy). *Let  $Y = \chi(\Omega)$  under the prior  $\Omega \sim \mu_\Delta$ . For every algorithm and every reference history  $H'_T$  independent of  $Y \sim \text{Unif}([K])$ ,*

$$\mathbb{P}\left(\bar{L}_Y(H'_T) \leq \frac{\Delta}{2}\right) \leq \frac{2}{K},$$

where  $\bar{L}_y(H'_T) = T^{-1} \sum_{t=1}^T \ell_y(A'_t)$  and  $A'_t$  denotes the arm selected in the reference history at time  $t$ .

*Proof.* For a fixed reference action sequence  $a'_1, \dots, a'_T$ , let

$$N'_y = \sum_{t=1}^T \mathbb{1}\{a'_t = y\}.$$

The average regret against the spike environment  $\omega^y$  is

$$\bar{L}_y(H'_T) = \Delta(1 - N'_y/T).$$

The event  $\bar{L}_y(H'_T) \leq \Delta/2$  implies  $N'_y \geq T/2$ . Since  $Y$  is uniform and independent of the reference sequence,

$$\mathbb{E}[N'_Y \mid H'_T] = \frac{1}{K} \sum_{y=1}^K N'_y = \frac{T}{K}.$$

Markov's inequality gives

$$\mathbb{P}(N'_Y \geq T/2 \mid H'_T) \leq \frac{2}{K}.$$

Averaging over  $H'_T$  proves the claim.  $\square$

**Lemma 6.3** (MAB information bound). *There is a universal constant  $c_0 > 0$  such that if  $\Delta^2 T/K \leq c_0$ , then every adaptive algorithm satisfies*

$$I(Y; H_T) \leq \frac{1}{4} \log(K/2),$$

where  $Y = \chi(\Omega)$  and  $\Omega \sim \mu_\Delta$ .

*Proof.* Let  $\mathbb{P}_0$  be the law of the history under the zero-mean reference model, let  $\mathbb{P}_j$  be the law under the spike environment  $\omega^j$ , and let

$$\bar{\mathbb{P}} = \frac{1}{K} \sum_{j=1}^K \mathbb{P}_j$$

be the marginal law of  $H_T$  under the prior  $\mu_\Delta$ . Under this prior,  $Y = j$  is equivalent to  $\Omega = \omega^j$ , but the mutual information is still the action-index information  $I(Y; H_T)$ , not information about an ambient full mean vector outside the prior support. Since  $\bar{\mathbb{P}} \geq K^{-1} \mathbb{P}_j$ , the likelihood ratio  $d\mathbb{P}_j/d\bar{\mathbb{P}}$  is bounded by  $K$ . The inequality

$$D_{\text{KL}}(P \parallel Q) \leq (2 + \log K) H^2(P, Q) \quad \text{whenever } Q \geq K^{-1} P,$$

together with the triangle inequality and convexity for squared Hellinger distance, gives

$$I(Y; H_T) = \frac{1}{K} \sum_{j=1}^K D_{\text{KL}}(\mathbb{P}_j \parallel \bar{\mathbb{P}}) \leq 4(2 + \log K) \frac{1}{K} \sum_{j=1}^K H^2(\mathbb{P}_j, \mathbb{P}_0).$$

Using  $H^2(P, Q) \leq D_{\text{KL}}(Q \parallel P)$  and the adaptive Gaussian KL chain rule under the zero model,

$$\frac{1}{K} \sum_{j=1}^K H^2(\mathbb{P}_j, \mathbb{P}_0) \leq \frac{1}{K} \sum_{j=1}^K D_{\text{KL}}(\mathbb{P}_0 \parallel \mathbb{P}_j) = \frac{\Delta^2}{2K} \mathbb{E}_0 \sum_{t=1}^T \sum_{j=1}^K \mathbb{1}\{A_t = j\} = \frac{\Delta^2 T}{2K}.$$

Therefore

$$I(Y; H_T) \leq 2(2 + \log K) \frac{\Delta^2 T}{K}.$$

Choosing, for example,

$$c_0 \leq \inf_{K \geq 3} \frac{\log(K/2)}{8(2 + \log K)}$$

which is a strictly positive universal constant, proves the displayed bound.  $\square$

**Theorem 6.4** (MAB lower bound). *For Gaussian  $K$ -armed bandits with unit noise variance, over any ambient mean-vector class  $\Omega_{\text{MAB}}$  that contains the spike instances  $\{\Delta e_j : j \in [K]\}$  used below,*

$$\mathfrak{R}_T^* \geq c\sqrt{KT}$$

*for a universal constant  $c > 0$  whenever  $T \geq K \geq 3$ . In particular, this applies to any bounded class such as  $[0, 1]^K$  once the universal constant in the choice of  $\Delta$  is small enough.*

*Proof.* Choose

$$\Delta = c_1 \sqrt{K/T}$$

with  $c_1$  small enough that Lemma 6.3 applies. Let  $r = \Delta/2$ . Lemma 6.2 gives  $p_r \leq 2/K$ , while Lemma 6.3 gives

$$T\bar{C} = I(Y; H_T) \leq \frac{1}{4} \log(K/2).$$

For  $c_1$  small enough, condition (55) holds. Theorem 5.1, applied uniformly over algorithms, gives a Bayes regret lower bound under the prior  $\mu_\Delta$  supported on the spike subfamily. Since this prior is supported inside the ambient model class  $\Omega_{\text{MAB}}$ , the minimax regret over  $\Omega_{\text{MAB}}$  is at least this Bayes risk. Hence

$$\mathfrak{R}_T^* \geq \frac{Tr}{2} = \frac{T\Delta}{4} = c\sqrt{KT}$$

for a universal constant  $c > 0$ .  $\square$

On the upper-bound side, the same distinction between the full model environment and the action index is important. An action-index AIR analysis uses  $Y = \chi(\Omega) = A^*(\Omega)$  as the information target, rather than the full environment  $\Omega$ . This avoids charging information for distinctions between environments that do not change the optimal action. A model-index E2D relaxation can pay a  $K$ -factor both through model estimation and through the DEC term, whereas the action-index AIR identity in Theorem 3.5 incurs only a  $\log K$  information-gain term and pays the  $K$ -factor through the AIR coefficient (Xu and Zeevi, 2025). This nearly matches the lower bound.

### 6.3 Linear bandits lower bound via optimal-action index and reference history

Consider stochastic linear bandits with action set

$$\mathcal{A} = \{a \in \mathbb{R}^d : \|a\|_2 \leq 1\}.$$

Let the ambient model class be the Euclidean unit parameter ball

$$\Omega_{\text{lin}} = \{\omega \in \mathbb{R}^d : \|\omega\|_2 \leq 1\},$$

and regard an environment  $\omega \in \Omega_{\text{lin}}$  as the full mean parameter vector. At time  $t$ , after choosing  $A_t \in \mathcal{A}$ , the learner observes

$$O_t = \langle \omega, A_t \rangle + \xi_t, \quad \xi_t \sim N(0, 1),$$

where the noises are independent over time and independent of the learner's external randomization. For  $\omega \neq 0$ , the unique optimal action over  $\mathcal{A}$  is

$$A^*(\omega) = \frac{\omega}{\|\omega\|_2}.$$

At  $\omega = 0$ , fix any deterministic tie-breaking action; this case is not used by the lower-bound prior. The information index is the optimal action

$$\chi(\omega) = A^*(\omega), \quad Y = \chi(\Omega),$$

not the full environment  $\Omega$ . Thus two environments that have the same optimal action are not distinguished by the index unless the prior support itself makes them distinguishable.

We prove the standard  $\Omega(d\sqrt{T})$  minimax lower bound through the quantile indexed information theorem. The proof uses a hypercube prior together with an adaptive information bound. The role of the hypercube variable below is only to code the finitely many optimal actions in the prior support. It should not be read as replacing the ambient environment  $\Omega$  by a model label. Note that for a non-Gaussian prior, one cannot condition on the realized adaptive design matrix and then apply the fixed-design Gaussian-channel capacity formula, because the design itself is a statistic of the past observations and may carry information about the unknown environment and hence about  $Y$ . [Chen et al. \(2024, Section 3.3\)](#) prove a closely related lower bound by a different route: they use the interactive Fano method with a Gaussian prior and a unit-sphere calculation. Both approaches show that action-indexed information arguments can recover the sharp linear-bandit order, whereas model-indexed relaxations such as DEC can lose factors when they charge for estimating more of the model than is needed to identify the optimal action.

Let

$$V \sim \text{Unif}(\{\pm 1\}^d), \quad \Omega = \Delta V.$$

Assume  $\Delta\sqrt{d} \leq 1$ , so that this prior is supported on  $\Omega_{\text{lin}}$ . On this prior support,

$$Y = \chi(\Omega) = A^*(\Omega) = \frac{V}{\sqrt{d}}. \tag{77}$$

The map  $v \mapsto v/\sqrt{d}$  is one-to-one on  $\{\pm 1\}^d$ . Hence, for this prior only,  $V$  may be used as a code for the action index  $Y$ , and

$$I(Y; H_T) = I(V; H_T).$$

For an index value  $y = v/\sqrt{d}$ , write  $\ell_v$  for the regret under the corresponding environment  $\omega = \Delta v$ . The one-step regret of action  $a \in \mathbb{R}^d$ ,  $\|a\|_2 \leq 1$ , is

$$\ell_v(a) = \Delta\sqrt{d} - \Delta\langle v, a \rangle.$$

Assume  $T \geq d^2$  and choose  $\Delta \leq 1/\sqrt{d}$ , so that  $\|\Omega\|_2 = \Delta\sqrt{d} \leq 1$ .

**Lemma 6.5** (Linear ghost entropy). *There is a universal constant  $c_{\text{gh}} > 0$  such that every deterministic reference action sequence  $a'_1, \dots, a'_T$ , with  $\|a'_t\|_2 \leq 1$ , satisfies*

$$\mathbb{P}_{V \sim \text{Unif}(\{\pm 1\}^d)} \left( \frac{1}{T} \sum_{t=1}^T (\Delta\sqrt{d} - \Delta\langle V, a'_t \rangle) \leq \frac{\Delta\sqrt{d}}{4} \right) \leq \exp(-c_{\text{gh}}d).$$

Consequently, the same bound holds after averaging over any random reference history  $H'_T$  whose induced action sequence is independent of the action index  $Y$ , equivalently independent of  $V$  under the hypercube prior.

*Proof.* Let

$$\bar{a} := \frac{1}{T} \sum_{t=1}^T a'_t.$$

Since  $\|\bar{a}\|_2 \leq T^{-1} \sum_t \|a'_t\|_2 \leq 1$ , the event in the display implies

$$\Delta\sqrt{d} - \Delta\langle V, \bar{a} \rangle \leq \frac{\Delta\sqrt{d}}{4},$$

or equivalently,

$$\langle V, \bar{a} \rangle \geq \frac{3\sqrt{d}}{4}.$$

The random variable  $\langle V, \bar{a} \rangle = \sum_{i=1}^d V_i \bar{a}_i$  is a centered Rademacher sum with variance proxy  $\|\bar{a}\|_2^2 \leq 1$ . Hoeffding's inequality therefore gives

$$\mathbb{P} \left( \langle V, \bar{a} \rangle \geq \frac{3\sqrt{d}}{4} \right) \leq \exp \left( -\frac{9d}{32} \right).$$

Thus the first claim holds, for example with  $c_{\text{gh}} = 9/32$ . If the reference action sequence is random but independent of  $Y$ , then because  $Y = V/\sqrt{d}$  is a bijective function of  $V$  on the prior support, the reference sequence is independent of  $V$ . Conditioning on the reference history and applying the deterministic bound gives the second claim.  $\square$

**Lemma 6.6** (Adaptive hypercube information capacity). *For the hypercube prior above, every possibly randomized adaptive algorithm satisfies*

$$I(Y; H_T) = I(V; H_T) \leq \Delta^2 T,$$

where  $H_T = (A_1, O_1, \dots, A_T, O_T)$ .

*Proof.* Since  $Y = V/\sqrt{d}$  is a bijective function of  $V$  on the hypercube support,  $I(Y; H_T) = I(V; H_T)$ . It remains to bound  $I(V; H_T)$ . Let  $U$  denote the learner's internal random seed, independent of  $V$ . Then

$$I(V; H_T) \leq I(V; H_T, U) = \mathbb{E}_U I(V; H_T \mid U).$$

It is therefore enough to prove the claim after conditioning on  $U$ , so that the algorithm is deterministic.

For  $v \in \{\pm 1\}^d$ , let  $\mathbb{P}_v$  denote the trajectory law when  $\Omega = \Delta v$ . We use the chain rule over coordinates:

$$I(V; H_T) = \sum_{i=1}^d I(V_i; H_T \mid V_1, \dots, V_{i-1}).$$

Fix a coordinate  $i$  and a prefix  $u = (v_1, \dots, v_{i-1})$ . Let  $\mathbb{P}_{u,+}$  and  $\mathbb{P}_{u,-}$  be the trajectory laws after averaging uniformly over the remaining coordinates  $V_{i+1}, \dots, V_d$ , conditional respectively on  $V_i = +1$  and  $V_i = -1$ . Then

$$I(V_i; H_T \mid V_{<i} = u) = D_{\text{JS}}(\mathbb{P}_{u,+}, \mathbb{P}_{u,-}),$$

where  $D_{\text{JS}}$  denotes Jensen–Shannon divergence with equal weights. For any two probability measures  $P, Q$ ,

$$D_{\text{JS}}(P, Q) \leq \frac{1}{4} \left( D_{\text{KL}}(P \parallel Q) + D_{\text{KL}}(Q \parallel P) \right).$$

Next couple the two mixtures by using the same suffix. If  $w \in \{\pm 1\}^{d-i}$  and

$$v^+(u, w) = (u, +1, w), \quad v^-(u, w) = (u, -1, w),$$

then joint convexity of relative entropy gives

$$D_{\text{KL}}(\mathbb{P}_{u,+} \| \mathbb{P}_{u,-}) \leq \mathbb{E}_w D_{\text{KL}}(\mathbb{P}_{v^+(u,w)} \| \mathbb{P}_{v^-(u,w)}),$$

and the same argument gives the analogous reverse inequality.

For two fixed vertices  $v^+$  and  $v^-$  that differ only in coordinate  $i$ , the adaptive KL chain rule gives

$$\begin{aligned} D_{\text{KL}}(\mathbb{P}_{v^+} \| \mathbb{P}_{v^-}) &= \frac{1}{2} \mathbb{E}_{v^+} \sum_{t=1}^T \left( \Delta \langle v^+ - v^-, A_t \rangle \right)^2 \\ &= 2\Delta^2 \mathbb{E}_{v^+} \sum_{t=1}^T A_{t,i}^2. \end{aligned}$$

Here the action kernel contributes no KL because, after conditioning on the learner's internal randomness, the algorithm uses the same decision rule under both environments; only the Gaussian observation kernel changes. Similarly,

$$D_{\text{KL}}(\mathbb{P}_{v^-} \| \mathbb{P}_{v^+}) = 2\Delta^2 \mathbb{E}_{v^-} \sum_{t=1}^T A_{t,i}^2.$$

Combining the preceding displays yields

$$\begin{aligned} I(V_i; H_T \mid V_{<i} = u) &\leq \frac{\Delta^2}{2} \mathbb{E}_w \left[ \mathbb{E}_{v^+(u,w)} \sum_{t=1}^T A_{t,i}^2 + \mathbb{E}_{v^-(u,w)} \sum_{t=1}^T A_{t,i}^2 \right] \\ &= \Delta^2 \mathbb{E} \left[ \sum_{t=1}^T A_{t,i}^2 \mid V_{<i} = u \right], \end{aligned}$$

where the final expectation is under the original hypercube prior and the algorithm's trajectory law. Averaging over  $V_{<i}$  and summing over  $i$  gives

$$\begin{aligned} I(V; H_T) &\leq \Delta^2 \mathbb{E} \sum_{t=1}^T \sum_{i=1}^d A_{t,i}^2 \\ &= \Delta^2 \mathbb{E} \sum_{t=1}^T \|A_t\|_2^2 \\ &\leq \Delta^2 T. \end{aligned}$$

This proves the claim. □

**Remark 6.7** (Why the fixed-design log-determinant argument is not used). *For a Gaussian prior on a full parameter vector, posterior conjugacy gives the adaptive information identity*

$$I(\Theta; H_T) = \frac{1}{2} \mathbb{E} \log \det \left( I + \Sigma_0^{1/2} \left( \sum_{t=1}^T A_t A_t^\top \right) \Sigma_0^{1/2} \right),$$

and this is bounded by a trace-constrained log determinant. For the hypercube prior, however, the posterior is not Gaussian and the realized design matrix is itself a statistic of the observations. Conditioning only on that realized design matrix controls the conditional observation information given the design; it does not by itself control the information carried by the adaptive design. Lemma 6.6 therefore uses a coordinatewise adaptive KL argument to control  $I(Y; H_T)$  directly.

**Theorem 6.8** (Linear bandit lower bound). *For stochastic linear bandits with action set  $\{a \in \mathbb{R}^d : \|a\|_2 \leq 1\}$ , unit Gaussian noise, and unit-ball parameter class  $\Omega_{\text{lin}} = \{\omega \in \mathbb{R}^d : \|\omega\|_2 \leq 1\}$ ,*

$$\mathfrak{R}_T^* \geq c d \sqrt{T}$$

for a universal constant  $c > 0$ , whenever  $T \geq d^2$ .

*Proof.* Assume first that  $d$  is larger than a sufficiently large universal constant  $d_0$  to be fixed below. Choose

$$\Delta = c_1 \sqrt{\frac{d}{T}},$$

where  $c_1 > 0$  is a sufficiently small universal constant. Since  $T \geq d^2$ , choosing  $c_1 \leq 1$  ensures

$$\Delta \sqrt{d} = c_1 \frac{d}{\sqrt{T}} \leq c_1 \leq 1,$$

so the hypercube prior is supported on the unit parameter ball.

Let

$$r = \frac{\Delta \sqrt{d}}{4}.$$

Lemma 6.5 gives the reference ghost-good probability bound

$$p_r \leq \exp(-c_{\text{gh}} d).$$

Lemma 6.6 gives the adaptive action-index information bound

$$T\bar{C} = I(Y; H_T) = I(V; H_T) \leq \Delta^2 T = c_1^2 d.$$

Choose  $c_1$  so small that  $c_1^2 \leq c_{\text{gh}}/4$ , and then choose  $d_0$  so large that  $\log 2 \leq c_{\text{gh}} d/4$  for all  $d \geq d_0$ . Then, for all  $d \geq d_0$ ,

$$T\bar{C} + \log 2 \leq \frac{1}{2} \log \frac{1}{p_r}.$$

Theorem 5.1, applied to the action index  $Y = \chi(\Omega) = A^*(\Omega)$ , gives a Bayes regret lower bound under the hypercube prior of at least

$$\frac{Tr}{2} = \frac{T\Delta\sqrt{d}}{8} = \frac{c_1}{8} d \sqrt{T}.$$

Since this prior is supported on  $\Omega_{\text{lin}}$ , the minimax regret over  $\Omega_{\text{lin}}$  is at least this Bayes risk.

It remains to handle the finitely many dimensions  $1 \leq d < d_0$ . For these dimensions, use the two-point prior  $\Omega = \sigma \delta e_1$ , where  $\sigma \sim \text{Unif}(\{\pm 1\})$ ,  $e_1$  is the first coordinate vector, and  $\delta = (2\sqrt{T})^{-1}$ . This prior is also supported on the unit parameter ball. Under  $\sigma = +1$ , the optimal action is  $e_1$ ; under  $\sigma = -1$ , the optimal action is  $-e_1$ . Let  $\mathbb{P}_+$  and  $\mathbb{P}_-$  be the laws of the history under these two environments, and let  $\mathbb{P}_{\pm}^{t-1}$  denote the law of the history before action  $t$ . For any algorithm,

$$\begin{aligned} \frac{1}{2} \mathbb{E}_+ [\delta(1 - A_{t,1})] + \frac{1}{2} \mathbb{E}_- [\delta(1 + A_{t,1})] &= \delta \left( 1 - \frac{1}{2} (\mathbb{E}_+ A_{t,1} - \mathbb{E}_- A_{t,1}) \right) \\ &\geq \delta (1 - \text{TV}(\mathbb{P}_+^{t-1}, \mathbb{P}_-^{t-1})). \end{aligned}$$



By Pinsker’s inequality and the adaptive Gaussian KL chain rule,

$$\text{TV}(\mathbb{P}_+^{t-1}, \mathbb{P}_-^{t-1}) \leq \sqrt{\frac{1}{2} D_{\text{KL}}(\mathbb{P}_+^{t-1} \parallel \mathbb{P}_-^{t-1})} \leq \delta \sqrt{t-1} \leq \frac{1}{2}.$$

Therefore each round has Bayes regret at least  $\delta/2$ , and the total Bayes regret is at least  $T\delta/2 = \sqrt{T}/4$ . Since  $d < d_0$ , this is at least  $(4d_0)^{-1}d\sqrt{T}$ . Reducing the universal constant  $c$ , if necessary, completes the proof for all  $d$ .  $\square$

## 6.4 Bellman-rank as indexed representation

Low Bellman rank was introduced by Jiang et al. (2017) as a sufficient structural condition under which contextual decision processes with suitable bilinear Bellman-error structure are PAC-learnable. In the present language, a Bellman-rank factorization can help build a low-dimensional Bellman-sufficient representation when it is supplemented by predictive, update, loss, and index sufficiency; the factorization alone controls Bellman-error evaluation and does not, by itself, determine the full observation process. The following proposition is structural; it is not a new sharp lower bound for every Bellman-rank class.

**Proposition 6.9** (Bellman rank with predictive sufficiency gives an indexed representation). *Suppose a contextual decision process admits an exact Bellman-error factorization of rank  $d$  in the sense of Jiang et al. (2017): for the stages and hypothesis-policy pairs in the planning class, the relevant Bellman error can be written as an inner product of two  $d$ -dimensional witness-feature vectors. Fix a prior  $\mu$  and an index map  $\chi$ . If the observation model is dominated by a common  $\sigma$ -finite measure and if, conditional on the current posterior or calibrated confidence object, the predictive law, the posterior-reference update, the relevant loss or benchmark terms, and the conditional  $\chi$ -indexed predictive laws and losses in Definition 2.3 are all determined by these Bellman-error features together with the retained planning variables, then the process admits an indexed Bellman sufficient representation with state that may be chosen as a posterior or calibrated confidence object over the  $d$ -dimensional witness coordinates, augmented by the planning variables needed for Bellman updates. Consequently, whenever one constructs a valid information supersolution and a valid ghost-good-mass supersolution for the induced feature experiment, for example in a linear-Gaussian or uniformly bounded-likelihood setting, Theorem 5.3 yields a Bellman quantile-index lower certificate.*

*Proof.* The Bellman-error factorization supplies  $d$ -dimensional coordinates sufficient to evaluate the one-step Bellman-error terms covered by the factorization. It does not by itself imply predictive sufficiency, loss sufficiency, update sufficiency, or index sufficiency. These are precisely the additional assumptions imposed in the proposition. If the predictive observation law and posterior update are functions of this statistic and the current posterior, and if the relevant losses, benchmarks, and  $\chi$ -conditional predictive objects are functions of the same retained state, then Definition 2.3 is satisfied with  $S_t$  equal to the posterior or calibrated confidence object over these features together with the accumulated planning state. The information and ghost-good Bellman recursions are then well-defined on the same state. Applying Theorem 5.3 to any valid pair of supersolutions for this state space gives the claimed certificate. The proposition is intentionally conditional because Bellman-rank learnability is primarily a structural upper-bound statement; sharp lower bounds require a packing or ghost-entropy calculation inside the particular rank- $d$  class.  $\square$

This application clarifies the connection between the present framework and the broader perspective on representation learning developed in Li and Xu (2026); Xu (2026). The point is not that

every low-Bellman-rank problem has the same minimax rate or that Bellman rank alone determines a lower bound. The point is that a Bellman-rank factorization can provide a candidate state on which the upper supersolution and the lower quantile-information certificate should be compared, after the necessary predictive, update, loss, and index-sufficiency conditions have been verified.

## 7 Conclusion

The paper identifies the statistical core of interactive decision making as an information-risk Bellman problem. A frequentist environment remains fixed; the proof chooses a Bellman-sufficient state, an information index, and a calibrated reference update. On that representation, the upper side is a log-penalized Bellman supersolution with coordinate potential  $\gamma \log(1/q_t(\chi(\omega^*)))$ , and the lower side is a Bellman-Fano ghost certificate whose entropy is  $-\log p_r$ . When these two logarithmic quantities close at the same radius, Theorem 4.4 gives a matching minimax rate. Equivalently, the critical balance is

$$T\bar{C}_\chi \asymp \log \frac{1}{p_r},$$

now interpreted on a controlled reference trajectory rather than on a fixed nonadaptive experiment.

AIR and MAIR are two choices of index within the same log-gain Bellman supersolution. Choosing  $\chi(\Omega) = A^*(\Omega)$  gives action-index AIR, which is often the right description of decision difficulty. Choosing  $\chi(\Omega) = \Omega$ , or the active reward vector in a bandit problem, gives model-index MAIR, which is often the convenient description of estimation geometry. With the fixed-truth coordinate bracket, the Bayesian posterior-averaged bracket, and the robust belief relaxation kept separate, the exact AIR/MAIR Bellman program, UCB, E2D/DEC, and AMS/EBO become four algorithmic forms of the same indexed log-potential identity: the exact program optimizes the dynamic information-potential Bellman recursion; UCB certifies the fixed-truth bracket through self-normalized calibration and optimism; E2D/DEC optimizes a one-step frequentist offset that drops or freezes the Bellman continuation; and AMS/EBO optimizes a KL-dual log-partition relaxation over candidate beliefs.

The kernel-bandit application illustrates why this separation matters. The finite-marginal indexed Bellman program is the exact benchmark on the sufficient state. GP-UCB is practical because it keeps the realized log-determinant information gain of the actual trajectory. AMS/EBO, in an AIR form, can scale with the entropy of the finite or low-dimensional optimal-action index and the sequential AIR bracket, provided the robust belief set is calibrated to the fixed truth. GP-E2D optimizes a one-round DEC offset, but its interaction with realized information gain is a theorem to be proved rather than a formal consequence of the offset definition; without localization a global RKHS DEC may be vacuous. Thus the right comparison is not a universal conversion coefficient. It is whether a single Bellman-sufficient state supports matching upper supersolutions and lower quantile certificates without charging estimation difficulty twice.

Several directions remain. The first is to identify representations for which the exact information-potential Bellman upper certificate and the Bellman-Fano lower certificate match up to constants or logarithms. The second is to replace one-step DEC calculations by dynamically consistent Bellman relaxations when posterior evolution matters. The third is to sharpen the finite-action kernel theory by proving which of the indexed Bellman DP, GP-UCB, AMS/EBO, or GP-E2D attains the finite-marginal information scale under the weakest calibration assumptions. These are representation-level questions. The goal is not a universal one-shot notion of dimension, but a distribution-dependent dynamic theory of state spaces that accounts for estimation, decision, and lower-bound entropy costs once, rather than repeatedly.

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